

# Currency risk and market integration

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# Currency risk and market integration

## Abstract

Market integration and currency risk are two main factors that distinguish international investment and financing decisions. Hence, we investigate the impact of currency factor on the dynamics of market integration. We compare integration indices estimated from conditional international asset pricing models with and without real exchange risk. We also examine the sensitivity of the Pukthuanthong and Roll (2009) R-square to omitted currency factors. In general, although the currency risk is significantly priced, it does not affect the level and the dynamics of the integration measure except under crises conditions.

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## Introduction

Market integration and currency risk are two main factors that distinguish international investment and financing decisions. They impact portfolio management, asset valuation, discount rates as well as cost of funding. Indeed, in a setting where capital markets are perfectly integrated and exchange risk is of no consequence, the global investment and financial decisions would be akin to domestic ones and models used in the domestic context would be applicable internationally.<sup>1</sup>

It is widely accepted that emerging markets (EMs) are not well integrated, currency risk is an important consideration in most markets at most of the time, and both are time varying. Overtime, the importance of EMs in global portfolios has increased substantially and is likely to grow in the future. Hence, in this paper, we examine the role of currency risk on time-varying integration of twenty emerging markets (EMs). Since a test and measure of market integration requires a specification of the underlying asset-pricing model, we base our investigation on two theoretical constructs. First, we use a dynamic version of the Chaieb and Errunza (2007, henceforth CE) equilibrium model that nests both market integration and currency risk. Second, we use the multi-factor model in the vein of Pukthuanthong and Roll (2009, henceforth PR). Specifically, we estimate the time-varying index of market integration of the EM stock indices using the two models with and without exchange risk.

Our paper contributes to the literature on market integration as well as currency risk. Some examples of tests of market integration include, Bekaert and Harvey (1995, henceforth BH), Carrieri, Errunza and Hogan (2007, henceforth CEH), Karolyi (2003), Baele (2005) Pukthuanthong and Roll (2009, henceforth PR), Bekaert, Harvey, Lundblad, and Siegel (2011), and Carrieri, Chaieb and (2013).<sup>2</sup>

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<sup>1</sup> Note that a perfectly integrated market would imply absence or ineffectiveness of explicit and implicit barriers to cross-border investment flows. Explicit barriers include legal restrictions on ownership, foreign exchange controls that are imposed by the governments of borrowing and creditor countries. Implicit barriers encompass risks related to political uncertainty, incomplete, inaccurate or asymmetric information, quality of governance, market size, illiquidity, and market regulation.

<sup>2</sup> BH were the first to study the time variation in the degree of integration for their sample of EMs. CEH investigate the time variation in market integration using the integration index of Errunza and Losq (1985, henceforth EL), and find strong evidence for time-varying integration and a significant price of conditional market risk. Karolyi (2003) estimates the time-varying EL index of market integration and finds similar results to CEH. Baele (2005)

While these studies shed light on the issue of time-variation in integration and its drivers, they do not explicitly recognize the role of currency risk.<sup>3</sup> The only exceptions are Choi and Rajan (1997) and Carrieri and Majerbi (2006). Choi and Rajan (1997) perform a joint test of market segmentation and exchange risk in the context of a multi-factor model with global and local risk factors on 337 individual stocks from seven major countries. They uncover significant currency risk premia. Using a similar approach, Carrieri and Majerbi (2006) uncover a significant unconditional exchange risk premium in nine emerging stock markets. Unfortunately, the unconditional setup in both papers does not allow the authors to account for the time variation in world market integration.

A number of papers based on international asset pricing models of Solnik (1974), Sercu (1980), Stulz (1981), and Adler and Dumas (1983) incorporate exchange risk in fully integrated markets. Some examples are, Dumas and Solnik (1995), De Santis and Gerard (1998), Vassalou (2000), Hardouvelis, Malliaropoulos and Priestley (2006), Phylaktis and Ravazzolo (2004), Carrieri, Errunza and Majerbi (2006), Moerman and van Dijk (2010), and Korajczyk and Viallet (1989). Their general conclusion is that the currency risk is conditionally priced, especially during crises periods. However, these studies do not investigate the level and dynamics of market integration since perfect market integration is a fundamental assumption of the underlying models. Indeed, to the best of our knowledge, no study combines both phenomena –market segmentation and currency risk - in a joint conditional setting.

Hence, we first use the CE model to investigate the relation between currency risk and the degree of market integration. According to the CE model, the expected return on a locally traded EM security should command a global market risk premium, global currency risk premiums and two local risk

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develops a volatility spillover model with regime switches in the spillover parameters and finds that Western European countries are highly integrated into the world market. PR use R-squares from the regression of a country's index returns on common global factors, to investigate trends in global integration. More recently, Bekaert, Harvey, Lundblad, and Siegel (2011) use valuation ratios to measure market segmentation and relate them to regulations with respect to foreign capital flows and other non-regulatory factors. Carrieri, Chaieb and Errunza (2013) study the relation between market integration and explicit and implicit barriers to cross-border portfolio flows.

<sup>3</sup> Some other studies do add the exchange risk factor to different empirical specifications. For example, De Santis, Thanyalakpark and Batten (2003) add a set of exchange rate related information variables to the EL model, whereas Hardouvelis, Malliaropoulos and Priestley (2006) and Adler and Qi (2003) use a version of the regime-switching model of BH but modify their asset pricing specification to include the currency risk.

premiums: A conditional market risk premium in the vein of EL and a segflation risk premium from bearing inflation risk in the presence of barriers.<sup>4</sup> The CE model nests the EL model when PPP holds and the Adler and Dumas (1983) model when markets are integrated. The comparison of the two integration measures based on the CE and EL models allows us to empirically assess the impact of omitted currency risk factors. We also derive conditions under which the two integration indices would be different. We show that the difference increases with the global currency premium and the sum of global currency and segflation premiums. It decreases in the covariance between the EM index's return and its diversification portfolio. The diversification portfolio is the projection of the EM index return on the span of the substitute assets that are globally accessible to investors.

We test conditional versions of the CE and EL models using the GARCH-M methodology of De Santis and Gerard (1998) for twenty MSCI EM indices over the period 1988-2010.<sup>5</sup> We focus on emerging markets given the ample evidence of full integration of the developed markets over our sample period. We find that global currency, segflation, global market and local market risks are priced factors. Despite the increase in integration across countries over time, emerging market's equity returns continue to depend strongly on local factors (Lewis, 2011). The average value of the total premium is close to the average value of the premium for global and local market risk. The average sum of the premiums for global currency risk and segflation risk is relatively small, and their contribution to total premium is significant only during crises periods. The two integration estimates are very similar most of the time except during currency and financial crises (such as the 1994 Peso problem, 1998 Russian default, 2007-2008 subprime crisis), when the global currency risk is nontrivial and the segflation risk is sizeable. Overall, despite the statistical and economic significance of the currency premiums, the currency risk factors do not significantly affect the integration measure.

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<sup>4</sup> The segflation risk is the covariance between the EM return index and inflation risk conditional on the returns on substitute assets that can be freely traded by investors of all nationality. With perfect substitutes for EM securities in the world market, the two conditional market and segflation risk premiums vanish.

We also address the potential issue of identification that might arise in case the foreign exchange crisis manifested by deep devaluations arise around the same time frame as the financial market liberalization, i.e. governments time the liberalization event around currency crisis. A time series analysis suggests that in most cases, the two phenomena do not seem to overlap, thus alleviating the concern. In addition, we test and confirm that the CE and EL integration indices are not related to changes in foreign exchange regimes for most countries. Robustness check on the parameterization of the prices of risk, where prices of global and local risk factors are assumed to be constant and the covariances vary through time, confirm our main findings. Our results are also robust with fewer instruments and different functional forms.

Next, we examine the role of currency risk factor based on a multifactor model. PR propose the R-square of a regression of returns on common variables as a measure of market integration. We examine the sensitivity of the R-squares based on common factors that include or omit the currency factors. We again investigate the same twenty EM markets over the period 1995-2008. In this case, we use the IFC/S&P emerging market database due to data limitation as explained in section III. We use the trade weighted major currency and emerging market currency indices defined in section I as well as bilateral exchange rates as exchange risk factors. Our main results carry through. The cross-country average R-squares with and without exchange risk are very similar although there are country specific variations during financial and currency crises periods. Again, the average bias between the estimates of integration measures with and without exchange risk is very small.

The rest of the paper is organized as follows. Section I presents the underlying equilibrium asset-pricing models, the empirical methodology, and data. Section II, discusses the results, identification issues and robustness checks. Section III presents the PR multi-factor model, the empirical methodology, and findings. Conclusion follows.

# I-The role of currency risk on market integration based on equilibrium asset pricing models

In this section, we examine whether systematic currency risk affects the level and dynamics of market integration. We estimate the integration measure from two equilibrium asset-pricing models, one that accounts for real currency risk and barriers to portfolio investments and one that accounts for barriers to portfolio investments but assumes purchasing power parity (PPP) holds and hence omits the currency risk factors.

## I-1 The underlying models

We implement two equilibrium international asset-pricing models (IAPMs). The IAPM of Chaieb and Errunza (2007) and the IAPM of Errunza and Losq (1985). The former accounts for capital inflow barriers to international investment and deviations from purchasing power parity. The latter assumes PPP holds. Both models assume a two-country world and two sets of securities. Domestic (e.g., the United States) securities can be traded by all investors. Foreign (e.g., the emerging market) securities can be held only by foreign investors.

According to CE, the equilibrium expected return on the foreign market portfolio is given by:

$$E_{t-1}^{CE}(r_{I,t}) = \delta_{W,t-1} cov_{t-1}(r_{I,t}, r_{W,t}) + \sum_{l=1}^{L-1} \delta_{l,t-1} cov_{t-1}(r_{DP,t}, q_{lt}) + \lambda_{I,t-1} var_{t-1}(r_{I,t} | \underline{r}_e) + \lambda_{q,t-1} cov_{t-1}(r_{I,t}, q_{It} | \underline{r}_e), \quad (1)$$

where  $r_{I,t}$  and  $r_{DP,t}$  are, respectively the excess returns on the EM portfolio and its diversification portfolio (DP), which is the projection of the EM portfolio on the space of substitute assets,  $r_{W,t}$  is the excess return on the world market portfolio,  $\underline{r}_e$  is the vector of returns on all substitute assets that can be bought by all investors irrespective of their nationality,  $E_{t-1}(\cdot)$  is the expectation operator conditional

on the information set at  $t-1$ ,  $\delta_{W,t-1}$  and  $\delta_{l,t-1}$  are, respectively, the prices of world market risk and global real currency risks,  $\lambda_{l,t-1}$  is the price of conditional market risk,  $var_{t-1}(r_{l,t} | \underline{r}_e)$ ,  $\lambda_{q,t-1}$  is the price of segflation risk,  $cov_{t-1}(r_{l,t}, q_{lt} | \underline{r}_e)$ ,  $q_{lt}$  is the change in real exchange rate of currency  $l$  in dollar terms, i.e.  $q_{lt} \equiv s_{lt} + \pi_{lt} - \pi_{0t}$ ;  $s_{lt}$  is the change in the nominal exchange rate of currency  $l$  in dollar terms;  $\pi_{lt}$  is the foreign inflation rate of country  $l$  measured in currency  $l$ ;  $\pi_{0t}$  is the US inflation rate measured in USD and assumed non-stochastic.<sup>6</sup> The rather significant inflation uncertainty for emerging markets does not allow us to proxy PPP deviations with nominal exchange rate changes as is the case for developed markets. For EMs, we measure PPP deviations with changes in real exchange rates. Using changes in the real exchange rates also help overcome empirical complications due to fixed exchange rate regimes or large discrete changes in nominal exchange rates due to sudden and sharp devaluations or peg removals.

If we constrain the prices of global real currency risk and of local segflation risk to be zero or assume PPP holds, we obtain the EL asset pricing equation,

$$E_{t-1}^{EL}(r_{l,t}) = \delta_{W,t-1} cov_{t-1}(r_{l,t}, r_{W,t}) + \lambda_{l,t-1} var_{t-1}(r_{l,t} | \underline{r}_e), \quad (2)$$

Estimation of the conditional version of the CE model delivers a time-varying market integration index,  $\Pi_t^{CE}$ , given by

$$\Pi_t^{CE} = \frac{[cov_{t-1}^{CE}(r_{l,t}, r_{DP,t})]^2}{var_{t-1}^{CE}(r_{l,t}) var_{t-1}^{CE}(r_{DP,t})}, \quad (3)$$

Though  $\Pi_t^{CE}$  is similarly defined as the EL integration index, denoted  $\Pi_t^{EL}$ , the estimates of the integration index depend on the specified asset-pricing model i.e. on the model's pricing kernel. The comparison of the two integration measures;  $\Pi_t^{CE}$  and  $\Pi_t^{EL}$ , allows us to assess the impact of omitted currency risk factors. First, we notionally assess whether currency risk omission leads to a systematic upward or downward bias in the integration measure. We then empirically assess the magnitude of the

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<sup>6</sup> See De Santis and Gerard (1998), Carrieri and Majerbi (2006), Carrieri, Errunza and Majerbi (2006), Chaieb and Errunza (2007) and Moerman and van Dijk (2010) for detail.



bias. We express the integration measure estimated from the EL model as a function of the estimate from the CE model and obtain the following inequality,<sup>7</sup>

$$\Pi_t^{EL} \leq \Phi_{t-1} + \Pi_t^{CE} (1 + \Psi_{t-1}) \quad (4)$$

where  $\Phi_{t-1}$  and  $\Psi_{t-1}$  are functions of the global currency premiums, the segflation premium and the second moments of the returns on the EM index and its diversification portfolio. They are, respectively, defined as:

$$\Phi_{t-1} \equiv \frac{[GCP_{t-1}(GCP_{t-1} + LCP_{t-1})]^2}{\sigma_{I,t-1}^{2,CE} \sigma_{DP,t-1}^{2,CE}} \quad (5)$$

$$\Psi_{t-1} \equiv \frac{2GCP_{t-1}(GCP_{t-1} + LCP_{t-1})}{cov_{t-1}^{CE}(r_{I,t}, r_{DP,t})} \quad (6)$$

where  $GCP_{t-1}$  is the global currency risk premium,  $LCP_{t-1}$  is the segflation risk premium that we also call local currency risk premium,  $\sigma_{I,t-1}^{2,CE}$ ,  $\sigma_{DP,t-1}^{2,CE}$  and  $cov_{t-1}^{CE}(r_{I,t}, r_{DP,t})$  are, respectively, the time-varying variances of the EM index, and of DP, and the time-varying covariance between the EM index and DP. For all second moments, the conditional mean is obtained from the CE model. While  $\Phi_{t-1}$  is positive, the sign of  $\Psi_{t-1}$  varies as a function of the sign of the currency premiums and the covariance between the EM index and DP.

**Proposition 1:** The two estimates of the CE and EL integration measures are almost equal at time  $t$  if the global currency premium is economically insignificant and the sum of global currency and segflation premiums is negligible.

(See Appendix A for proof of proposition 1 and Equation (4))

The omission of the real currency risk factor leads to a systematic upward bias on the estimated variances of the local market portfolio and its DP (see Appendix A Equations A.21 and A.22). However, the bias on the covariance between the local market portfolio and its DP could be positive or negative depending on the sign of the global and total currency risk premia. Indeed, currency risk premia can be either

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<sup>7</sup> Note that the sources of misspecification, such as errors in the specification of the dynamics of the covariance matrix and/or the dynamics of the prices of risk as well as the bias on the estimated risk premia from omitted factors, are not considered.

positive or negative because of the symmetric nature of exchange risk. Thus, the omission of currency factors could result in a higher or lower level of integration. Next, we empirically investigate whether the two estimates are economically significantly different.

## I-2 Empirical methodology

Our measures of the global real currency exposures of the EM index are the covariances between the diversification portfolio return and two currency indices (see also Ferson and Harvey (1993), Harvey (1995) and Carrieri, Errunza and Majerbi (2006) for the use of currency indices). The two currency indices are the major currency index (termed the MJ currency index) and the Other Important Trading Partner currency index (termed the EM currency index). These are the trade-weighted values of US dollar against a number of currencies where the trade-weights are allowed to vary over time. The MJ currency index includes sixteen currencies until the introduction of the euro in January 1999. After that, the index becomes a seven-currency index. The EM currency index includes mainly emerging market currencies. We take the inverse of the real currency indices so that the higher index values represent an appreciation of the foreign currency.

We estimate a conditional version of the CE model where we allow prices and quantities of risk to change through time.<sup>8</sup> We follow Bekaert and Harvey (1995) and estimate the model in two stages. In the first stage, the world market risk and the two global real currency risk prices are estimated from the following system of equations,

$$(7) \begin{cases} r_{W,t} = \delta_{W,t-1} h_{W,W,t} + \delta_{MJ,t-1} h_{W,q_{MJ},t} + \delta_{EM,t-1} h_{W,q_{EM},t} + \varepsilon_{W,t} \\ q_{MJ,t} = \delta_{W,t-1} h_{W,q_{MJ},t} + \delta_{MJ,t-1} h_{q_{MJ},q_{MJ},t} + \delta_{EM,t-1} h_{q_{MJ},q_{EM},t} + \varepsilon_{q_{MJ},t} \\ q_{EM,t} = \delta_{W,t-1} h_{W,q_{EM},t} + \delta_{MJ,t-1} h_{q_{MJ},q_{EM},t} + \delta_{EM,t-1} h_{q_{EM},q_{EM},t} + \varepsilon_{q_{EM},t} \\ \varepsilon_t | \Omega_{t-1} \sim N(0, H_t) \end{cases}$$

where  $\Omega_{t-1}$  is the set of information available at time  $t - 1$ , and  $h_{i,j,t}$  are the elements of  $H_t$ , the  $(3 \times 3)$  conditional covariance matrix of the 3 assets in the system conditional on time  $t$ .

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<sup>8</sup> Many previous studies show that it is necessary to allow the prices of risk to vary over time, see among others Harvey (1991), Ferson and Harvey (1991), Dumas and Solnik (1995), De Santis and Gerard (1997).

The second stage estimates the model country by country, conditioning on the estimates from the first stage. This approach yields common prices of world market risk and of global real currency risks across all assets consistent with the model prediction.

We further parameterize  $var_{t-1}(r_{I,t} | \underline{r_e})$  as  $h_{I,I,t} \left(1 - \frac{h_{I,DP,t}^2}{h_{I,I,t}h_{DP,DP,t}}\right)$ , with  $h_{I,DP,t}$  the time-varying covariance,  $h_{I,I,t}$  and  $h_{DP,DP,t}$  the time-varying variances. Conditional on time, different parameterizations of the conditional variance are equivalent only if the coefficient on the diversification portfolio is conditionally equal to one. We use the correlation based parameterization because it ensures that the integration measure, defined as one minus the ratio of the conditional variance to the total variance, is bounded at every point in time  $t$  by 0 and 1 (See Carrieri, Chaieb and Errunza, 2013 for a more extended explanation). Similarly,  $cov_{t-1}(r_{I,t}, q_{I,t} | \underline{r_e})$  is parameterized as  $h_{I,qI,t} \left(1 - \frac{h_{DP,qI,t}}{h_{I,qI,t}}\right)$ .

Hence, in the second stage and for each country, we estimate the following system of equations:

$$(8) \begin{cases} r_{I,t} = \hat{\delta}_{W,t-1}h_{I,W,t} + \hat{\delta}_{MJ,t-1}h_{DP,qMJ,t} + \hat{\delta}_{EM,t-1}h_{DP,qEM,t} + \lambda_{I,t-1}h_{I,I,t} \left(1 - \frac{h_{I,DP,t}^2}{h_{I,I,t}h_{DP,DP,t}}\right) \\ \quad + \lambda_{q,t-1}h_{I,qI,t} \left(1 - \frac{h_{DP,qI,t}}{h_{I,qI,t}}\right) + \varepsilon_{I,t} \\ r_{DP,t} = \hat{\delta}_{W,t-1}h_{DP,W,t} + \hat{\delta}_{MJ,t-1}h_{DP,qMJ,t} + \hat{\delta}_{EM,t-1}h_{DP,qEM,t} + \varepsilon_{DP,t} \\ q_{I,t} = \hat{\delta}_{W,t-1}h_{qI,W,t} + \hat{\delta}_{MJ,t-1}h_{qI,qMJ,t} + \hat{\delta}_{EM,t-1}h_{qI,qEM,t} + \varepsilon_{qI,t} \end{cases}$$

where  $\hat{\delta}_{W,t-1}$ ,  $\hat{\delta}_{MJ,t-1}$ , and  $\hat{\delta}_{EM,t-1}$  are the first-stage estimates of the prices of the world market, the MJ real currency, and the EM real currency risk factors respectively.

The vector of residuals,  $\varepsilon_t$ , obtains by stacking the residuals of the three equations of system (8) and the three first-stage estimated residuals,  $\hat{\varepsilon}_{W,t}$ ,  $\hat{\varepsilon}_{qMJ,t}$ , and  $\hat{\varepsilon}_{qEM,t}$  of system (7). That is,  $\varepsilon_t = (\varepsilon_{I,t} \ \varepsilon_{DP,t} \ \varepsilon_{qI,t} \ \hat{\varepsilon}_{W,t} \ \hat{\varepsilon}_{qMJ,t} \ \hat{\varepsilon}_{qEM,t})'$ . We assume  $\varepsilon_t | \Omega_{t-1} \sim N(0, H_t)$ , where  $H_t$  is the  $(6 \times 6)$  conditional covariance matrix of the 6 assets conditional on time  $t$  and on the estimated residuals from first stage.

The CE model implies that the price of world market risk must be positive. Furthermore, the model restricts the price of conditional market risk to be positive because the aggregate world risk aversion is

lower than the aggregate risk aversion of the  $i$ th emerging market. Thus, we parameterize the prices of risk as follows,

$$\begin{aligned}\delta_{W,t-1} &= (k'_w Z_{W,t-1})^2, \\ \lambda_{I,t-1} &= (k'_i Z_{t-1})^2,\end{aligned}$$

The prices of global currency risks and segflation risk factors can be negative and are parameterized as follows,

$$\begin{aligned}\delta_{MJ,t-1} &= k'_{mj} Z_{W,t-1}, \\ \delta_{EM,t-1} &= k'_{em} Z_{W,t-1}, \\ \lambda_{q,t-1} &= k'_e Z_{t-1},\end{aligned}$$

where  $Z_{t-1}$  is the set of time-varying global ( $Z_{W,t-1}$ ) and local ( $Z_{I,t-1}$ ) information variables. Note that the prices of global (market and real currency) risk factors are functions of global instruments, while the prices of conditional market and segflation risk factors are functions of global as well as local instruments. If local risk is priced, we should reject the hypothesis that  $k_i$  and  $k_e$  are jointly equal to zero.

As in De Santis and Gerard (1997), we adopt the diagonal representation of the multivariate GARCH model of Bollerslev, Engle, and Wooldridge (1988), which assumes that the variances in  $H_t$  depend only on past squared residuals and an autoregressive component, while the covariances depend on the past cross-product of residuals and an autoregressive component.<sup>9</sup> We also impose Ding and Engle's (2001) condition that assumes the process to be covariance stationary. The advantage of this multivariate GARCH in mean parameterization is that it ensures positive definiteness of the covariance matrix  $H_t$  while reducing the number of parameters to be estimated. The dynamics of the conditional second moment  $H_t$  are specified as

$$H_t = H_0 * (ii' - aa' - bb') + aa' * \varepsilon_{t-1} \varepsilon'_{t-1} + bb' * H_{t-1}, \quad (9)$$

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<sup>9</sup> De Santis and Gerard (1997, 1998) find weak evidence for volatility spillover with returns at the monthly frequency. Also, CEH find little evidence that world level shocks impact conditional variances and covariances of the other assets. More importantly, they show that the integration index measure is robust to the modeling of volatility spillover.

where  $i$  is a  $(6 \times 1)$  vector of ones,  $a$  and  $b$  are  $(6 \times 1)$  vectors of parameters, and  $*$  denotes the Hadamard (element by element) matrix product. We estimate the model by the quasi-maximum likelihood (QML) of Bollerslev and Wooldridge (1992), using the BFGS (Shanno 1985) algorithm to update the Hessian. The EL model obtains if currency risk is not priced or PPP holds. Keeping the dynamic specification for the conditional variance-covariance matrix  $H_t$  the same as before, the model in System (8) collapses then to a trivariate system as follows:

$$(8') \begin{cases} r_{I,t} = \delta_{W,t-1} h_{I,W,t} + \lambda_{I,t-1} h_{I,I,t} \left( 1 - \frac{h_{I,DP,t}^2}{h_{I,I,t} h_{DP,DP,t}} \right) + \varepsilon_{I,t} \\ r_{DP,t} = \delta_{W,t-1} h_{DP,W,t} + \varepsilon_{DP,t} \\ r_{W,t} = \delta_{W,t-1} h_{W,W,t} + \varepsilon_{W,t} \end{cases}$$

### I-3 Data

Our dataset includes the test asset returns and exchange rate factors, the substitute assets to construct the diversification portfolios (DPs), and the instrumental variables.

#### Asset returns

Data on monthly returns on MSCI EM indices are obtained from Datastream. The world market index is the MSCI world index. The set of emerging markets includes twenty emerging markets: Argentina, Brazil, Chile, China, Colombia, Czech Republic, Hungary, India, Indonesia, Jordan, Korea, Malaysia, Mexico, Pakistan, Peru, Philippines, South Africa, Taiwan, Thailand, and Turkey. The MSCI indices are market value weighted total returns expressed in U.S. dollar terms. The sample period is from January 1988 or later to December 2010. For the conditionally risk-free asset, we use the return on the one-month Eurodollar deposit. We compute the monthly excess returns by subtracting the Eurodollar rate from the monthly return on each security. Real bilateral exchange rates with respect to the dollar are computed on a price-adjusted basis using CPI indices available from the International Finance Statistics (IFS) database. Data on the real exchange rate indices (MJ and EM) is from the Federal Reserve Board. Table 1 reports the average and volatility of the emerging market returns and changes in bilateral real exchange rates. The average equity risk premium is positive for all markets except China and Jordan. The Turkish market is the most volatile over the full sample period and across the 20

emerging markets. Most currencies show a positive real appreciation on average. The changes in bilateral real currencies,  $q_t$ , display high volatility but still lower than the equity market volatility. The Argentinean peso is the most volatile currency in real terms.

[Insert Table 1 about here]

#### The substitute assets:

The substitute assets include the MSCI World index, thirty-four global market industries constructed according to the FTSE industry classification, seventeen closed-end country funds (CFs), eighty-four cross-listings whether these are direct placements, American depository receipts (ADRs), or global depository receipts (GDRs), and eleven country exchange traded funds (ETFs).<sup>10</sup> The World index and the global industries consist of only the developed markets. The stocks cross-listed outside the United States are listed in either the United Kingdom or Germany. To preserve the degrees of freedom in the regression, we include for each country only the first incepted country fund, the five earliest cross-listings when available and the first incepted country ETF. The monthly returns (adjusted for dividends) for CFs and ETFs are obtained from the Center for Research in Security Prices (CRSP) database. The return data on ADRs are collected from CRSP, while return data on GDRs are compiled from Datastream. A complete list of the set of substitute assets is reported in Appendix B. The diversification portfolios are constructed in a similar way as in CEH allowing the weights to vary upon the availability of new securities.

Table 1 also reports the pairwise correlations between each EM index return and the following three assets' returns: (1) the change in USD/EM real exchange rate, (2) the diversification portfolio of the EM index, (3) the world index return. The last column displays the correlation between the country diversification portfolio and the world index return. We observe that country index returns are on average positively correlated with real appreciation of their currencies vs. the US dollar. Also and as

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<sup>10</sup> Carrieri, Chaieb and Errunza (2013) show that the integration measures are robust to further augmenting the set of substitute assets with regional funds as well as different approaches in constructing the DPs.

expected, the countries with few substitute assets, such as Jordan, show low correlations with their diversification portfolios, while countries with significant substitute assets, such as Mexico, show high correlations with their diversification portfolios.

### Instrumental variables

In selecting the data on the global and local instrumental variables, we follow previous research to facilitate comparison.<sup>11</sup> The global information set includes the world dividend yield in excess of the risk free rate, the change in the US term premium, measured by the yield difference between the three-month T-bill and the 10-year T-bond, the US default premium measured by the yield difference between Moody's Baa and Aaa rated bonds, and the option volatility index (VIX).<sup>12</sup> The local information variables include the local equity return in excess of the risk free rate, the local excess dividend yield and the change in nominal bilateral exchange rate. The global set of instruments is used to condition the price of global market risk and the price of global real currency risk. The global and local instruments are used to condition the price of conditional market risk and segflation risk. All the information variables are one-month lagged. Since these instrumental variables have been widely used in other studies, we omit a detailed description of their properties. The online appendix Table A.1 show some basic statistics of the instruments, their pairwise correlations, and additional statistics on the asset returns.

## **II- Empirical Results based on equilibrium asset pricing models**

In this section, we discuss the empirical results from the estimation of the CE and EL models. Specifically, we examine the significance of the currency risk premia, which is the source of bias in the

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<sup>11</sup> See among others Ferson and Harvey (1993, 1994), Dumas and Solnik (1995), De Santis and Gerard (1998).

<sup>12</sup> The VIX data start on 1990. We appended the series with data on VXO from 1988 to 1990. See Carr and Wu (2006) for a description and comparison of the two indices. Essentially, they show that the two indices are highly correlated.

estimation of the integration measure. We also discuss identification issues and some robustness experiments.

## **II.1 Main Findings**

Since the results from the estimation of the prices of risk and diagnostic tests are similar to Chaieb and Errunza (2007) we just summarize the key findings. The price of world market risk is significant and is time-varying. Its average of 2.82 is economically reasonable and consistent with previous studies. The price of MJ real currency risk is not significant, while the price of EM real currency risk is marginally significant. Nevertheless, the average prices of MJ and EM real currency risks of -3.31, and 12.01, respectively, are economically meaningful. The price of conditional market risk is significant for all countries except China and is significantly time-varying in 16 cases. In 14 of the 20 countries, the segflation risk is significantly priced and is significantly time-varying in 12 cases. Finally, the joint test on the constant price of conditional market and segflation risk factors is rejected in all cases. Hence there is substantial evidence that emerging markets' equity returns depend upon both local and global risk factors. Online appendix Table A.2 reports the parameter estimates, specification tests and diagnostic tests from the two-step estimations of the CE model.

From the EL model estimation we obtain a significantly time-varying price of world market risk. Its mean of 2.6 is of similar magnitude as the one obtained from the CE model. The prices of conditional market risk factors are significant in all cases, are time-varying in 17 cases and of similar size as the ones obtained from the CE model's estimation.

Next, we examine the economic magnitude of the currency risk premium, which is relevant to assess the bias in the integration measure from omitted currency factor. We measure the premium for the EM indices associated with each risk factor as the product of the estimated price of risk and conditional covariance. The currency risk premium (CRP) is the sum of the global real currency and the local segflation risk premiums and is given by:



$$CRP_{I,t} = \sum_{l=MJ,EM} \delta_{l,t-1} h_{DP,q_l,t} + \lambda_{q,t-1} h_{I,q_l,t} \left( 1 - \frac{h_{DP,q_l,t}}{h_{I,q_l,t}} \right) \quad (10)$$

The market risk premium (MRP) is the sum of the world market and conditional market risk premiums and is given by:

$$MRP_{I,t} = \delta_{W,t-1} h_{I,W,t} + \lambda_{I,t-1} h_{I,t} \left( 1 - \frac{h_{I,DP,t}^2}{h_{I,t} h_{DP,t}} \right). \quad (11)$$

The total risk premium (TRP) is the sum of CRP and MRP and is given by:

$$\begin{aligned} TRP_{I,t} = & \delta_{W,t-1} h_{I,W,t} + \sum_{l=MJ,EM} \delta_{l,t-1} h_{DP,q_l,t} \\ & + \lambda_{I,t-1} h_{I,t} \left( 1 - \frac{h_{I,DP,t}^2}{h_{I,t} h_{DP,t}} \right) + \lambda_{q,t-1} h_{I,q_l,t} \left( 1 - \frac{h_{DP,q_l,t}}{h_{I,q_l,t}} \right) \end{aligned} \quad (12)$$

[Insert Table 2 about here]

The contribution of MRP and CRP to the total risk premium is displayed in Table 2. The contributions are measured by the ratio of the average (market or currency) premium in absolute terms to the sum of all the average risk premiums in absolute terms. In many cases, the average value of the total premium is close to the average value of the premium for global and local market risk, whereas the average sum of the premiums for global currency risk and segflation risk is relatively smaller. The contribution of CRP to total premium ranges from 5% in Thailand to 53% in Brazil. On average, the contribution of the global currency risk premium (GCP) to the total premium ranges between 1% for Jordan or Pakistan to 25% for China. The average contribution of the local currency risk premium (LCP) exceeds that of global currency risk premium for many countries. Additionally, there is important variation in the risk premiums through time and across countries. To further gauge the importance of the local currency risk premium, global currency risk premium, and total currency risk premium and their dynamics, we average across countries the different risk premiums. Figure 1 plots these time-varying premiums. The plot confirms that the global currency premium is overall small. Nevertheless, the contribution of the global currency risk premium to total premium is significant in crises periods.

Specifically, we observe large and negative global currency risk premia during the Asian and the Russian crises of 1997-1998 and a large and positive global currency premium over the recent financial crisis 2008-2009. The segmentation premium is large and negative in crises periods and it explains most of the total currency premium.

[Insert Figure 1 about here]

Table 3 reports statistics on the estimated integration indices from the CE and EL models, as well as on  $\Phi_{t-1}$  and  $\Psi_{t-1}$ . The mean integration indices from the CE model (EL model) range from 0.08 (0.07) to 0.95 (0.95). Mexico is the most integrated, while Jordan is the least integrated in our sample. The averages of  $\Pi_t^{CE}$  and  $\Pi_t^{EL}$  are very close. This is because  $\Phi_{t-1}$  and  $\Psi_{t-1}$  are on average zero. The average bias between the EL and CE integration measures does not exceed 5%. In six cases, there is no bias on average. In eight cases, the bias is negative meaning that omission of currency risk factors produce slightly lower integration indices on average.

[Insert Table 3 about here]

Figure 2 summarizes the estimated integration indices. We present them as equally weighted averages at each point in time from the two models. The average estimates are almost identical. While in general the two estimates at the individual country level are not sensibly different because of a trivial global currency risk premium in non-crisis periods, some of the variability observed in  $\Pi_t^{EL}$  estimates seems to be driven by omitted currency risk factors. For instance, the sudden increase in the level of integration by late 2008 for Jordan is removed when currency risk is added. For Brazil, on 07/1997 we observe a drop in the  $\Pi_t^{EL}$  while there is no such drop in  $\Pi_t^{CE}$ . A similar decrease in  $\Pi_t^{EL}$  is noted around the recent financial crisis while  $\Pi_t^{CE}$  remained steady at around 0.85 from 2000 till 2010. It is also noteworthy that Argentina, China, Colombia, Czech Rep., Malaysia, and Philippines experienced a drop in their level of integration following the subprime crisis. The individual country level plots of the CE and EL integration indices are available from the authors.

[Insert Figure 2 about here]

## II-2 Identification issue

Next, we address the potential identification issue that might arise since many of our sample countries had fixed exchange rate regimes during our study period that experienced foreign exchange and balance of payments crises (e.g. Mexico 1994, Thailand 1997, Malaysia 1998, etc.). It is possible that some of these crises manifested by deep devaluations may have occurred around the same time frame as the financial market liberalization, i.e. governments may time the liberalization event around currency crisis. To investigate this issue, for each EM, we plot the time series of nominal exchange rate, the integration index and the often-used liberalization proxy of percentage of market capitalization available for investment to foreign investors. In Figure 3, we report plots of EMs that have experienced deep devaluations. The plots suggest that in most cases, the two phenomena do not seem to overlap, thus alleviating the identification concern. For instance, the level of integration of the Mexican market remain high at around 0.9 after the sharp devaluation of the Mexican peso in 1994. From July 1997 to January 1998, Indonesian rupiah has depreciated by over 75% with little impact on the upward trend in the investable market capitalization or the level of market integration.

[Insert Figure 3 about here]

## II-3 Relation between integration indices and foreign exchange regimes

We test whether the CE and EL integration indices are related to changes in foreign exchange regimes by running similar regressions as in BH. We regress nominal exchange rate changes on the CE (EL) estimated measures of market integration, as well as lagged nominal changes in exchange rates and the Eurodollar deposit rate. Table 4 presents the coefficient on the estimated integration,  $\alpha_3$ , the heteroskedasticity-consistent  $\chi^2(1)$  and p-value associated with  $\alpha_3$ , as well as the difference between the adjusted  $R^2$ 's of the following regression models:

$$s_{l,t} = \alpha_0 + \alpha_1 s_{l,t-1} + \alpha_2 i_{t-1} + \alpha_3 \hat{\Pi}_{t-1}^M + u_t \quad (13a)$$

$$s_{l,t} = \alpha_0^* + \alpha_1^* s_{l,t-1} + \alpha_2^* i_{t-1} + u_t^* \quad (13b)$$

where  $i_t$  is the Eurodollar deposit rate, and  $\hat{\Pi}_t^M$  is the estimated degree of integration from model  $M = CE, EL$ .

[Insert Table 4 about here]

For both CE and EL models, only three of 20 countries show significant coefficient  $\alpha_3$  at the 5 percent level. The set of countries with significant  $\alpha_3$  includes Brazil, Philippines, and Turkey, when  $\hat{\Pi}_t^M$  is estimated from CE. This set includes Argentina, Brazil, and Philippines, when  $\hat{\Pi}_t^M$  is estimated from EL. Additionally, the increases in the adjusted  $R^2$  are very small for all countries and for the two models. We redo the regressions (13a) and (13b) replacing  $s_{l,t}$ , the  $l$  country's change in nominal exchange rates, by  $q_{l,t}$  the  $l$  country's change in real exchange rates. Results reported on the online appendix table A.3 show that of the 20 countries, only Chile, Pakistan and Philippines (Chile) display significant coefficients on the CE (EL) estimated integration indices at the 5 percent level. The increases in the adjusted  $R^2$  are again marginal. Hence, these diagnostics confirm the weak relation between exchange rate changes and the integration measures from the CE and EL models. Note that BH also do not find strong evidence that changes in nominal exchange rates and their integration measure are interrelated.

## II-4 Robustness

Since the parameterization of the prices of risk could be a source of estimation error, we analyzed a version of the CE and EL models in which the prices of global and local risk factors are assumed to be constant, whereas the covariances vary through time (Results available from the authors). The results are unchanged. Although we assume constant prices, we uncover that the world market risk and EM real currency risks are priced. Also the conditional market and segflation risks are significantly priced for many countries. The magnitudes of the unconditional prices of risk are very close to the average prices of risk estimates. More importantly, our main finding that the global currency risk and segflation risk

factors do not significantly affect the integration measure is confirmed. The specification tests reported on the online appendix table A.2 show that the null of constant prices is rejected and hence it is important to account for the dynamics of the prices of risk. Additional robustness checks on the parameterization of the prices of risk with fewer instruments and different functional forms confirm our main findings. (Results are available from the authors.)

### **III- The role of currency risk on market integration based on a multifactor-model**

In this section, we examine whether currency risk factors affect the level and dynamics of market integration based on multi-factor models. We estimate the PR R-squares integration measure from two multi-factor models; one that accounts for currency risk factors and one that omits the currency risk factors. We next discuss the underlying models, the empirical methodology and results.

#### **III-1 The model**

Ross (1976) suggests that common variables that explain contemporaneously a wide cross-section of returns should be priced. The empirical challenge of the APT model is the choice of the common factors. These could be either pre-specified observable factors or estimated from principal component analysis. Ferson and Harvey (1993) investigate the APT model in which the pre-specified factors include the world market portfolio, an aggregate of exchange rate, and other global economic risk factors. They find that exchange rates are sources of common variation and are priced. Choi and Rajan (1997) examine the role of exchange risk by estimating a two-factor model that includes global and domestic market risk factors and a three-factor model that augments the two-factor model with an exchange risk factor.

PR propose the R-square of a regression of returns on common global variables as a measure of market integration. Each EM's stock market return follows a K-factor structure,

$$R_{jt} = \alpha_j + \sum_{k=1}^K \beta_{kj} f_{k,t} + e_{jt} \quad (14)$$

where  $R_{jt}$  is the return of country  $j$ 's broad market index,  $\beta_{kj}$  is the  $k$ 's factor loading, and  $f_{k,t}$  is the  $k$ th common global factor at time  $t$ ,  $k=1, \dots, K$ . The adjusted R-square from the multi-factor model (14) is their measure of integration. PR argue that “if something is omitted, it is omitted for all countries and might not have much impact on the pattern of relative measures of market integration”. They also suggest “there could be a relative bias if an omitted factor is singularly important for a particular country”.

### III-2 Empirical methodology

We examine sensitivity of the PR R-square to omitted currency factors. Because MSCI EM indices are available at the weekly frequency from 2000, we instead use the IFC Investable indices of S&P, which are available weekly since 1989 and span the period 1989-2008. It also allows us to investigate whether the currency risk plays an important role for a different dataset. The real currency indices are available from the FED at the monthly frequency only. This is because the CPIs are not available at higher frequency. We then use the weekly nominal MJ and EM currency indices of the FED. The MJ currency index is available weekly from the start of our sample period, while the EM currency index is available at the weekly frequency starting on 1995. The bilateral real exchange rates are computed from the weekly nominal rates and the monthly CPIs. We estimate the R-squares over the period 1995-2008 using alternatively nominal and real bilateral exchange rates. Since results are very similar, we only report and discuss estimation with real bilateral rates.

We estimate the adjusted R-squares in the vein of the PR procedure in two ways. Approach I first estimates principal components from the global industries. We then add the countries' substitute assets as well as exchange rates in the regressions of country index returns. This approach follows the insights of the EL model. It focuses on the marginal impact of substitute assets and exchange rate factors, given a common set of global variables. Ideally, we would include all countries substitute assets and exchange rate factors but this is empirically unfeasible. Thus, a shortcoming of this approach is that it assumes

that the loading of countries' index returns on the substitute assets and exchange rates of the other countries is marginal. Approach II uses the global industries, substitute assets (CFs, ADRs, ETFs) from all sample markets that are present every year over 1994-2008, as well as the exchange rates. Approach II is more in line with the APT and PR. Common factors are first extracted and then used for all asset returns. However, this approach would result in lower integration measures compared to CE and EL integration indices as a total of 86 substitute assets did not enter the regressions because they don't span the full period 1994-2008.

Both approaches are applied in two steps. For approach I, we first estimate the principal components in year  $t$  using the eigenvectors computed with the  $10 \times 10$  covariance matrix of the level 2 ten global industry returns from year  $t-1$ . We retain the first three out-of-sample principal components, which explain more than 85% of the total volatility in all of the 14 years. In the second step, we use weekly returns with a one-year estimation window and run the rolling non-overlapping regressions. To enter the regression, the countries must have at least 30 valid weekly IFCI returns during the year. The regressors include the three out-of-sample principal components and the country specific set of substitute assets used for the diversification portfolios detailed in Appendix B. We then obtain the adjusted R-squares,  $R2\_noFX$ . We redo step (2) including MJ, EM as well as bilateral currency factors and obtain the adjusted R-squares,  $R2$ .

For approach II, we estimate the global factors from the ten global industries as well as all of the substitute assets (CFs, ADRs, ETFs) for all EMs that are present every year over 1994-2008. The dimension of the covariance matrix in this case is  $41 \times 41$ . We retain the first 10 out-of-sample principal components, which account for about 75% of the cumulative eigenvalues across the 14 years. We compute the adjusted R-squares,  $R2\_noFX$ , from the non-overlapping rolling regressions with a one-year estimation window of country returns on the 10 out-of-sample principal components. To examine the impact of omitted currency factors, we re-estimate the global factors by augmenting the set with the

MJ nominal currency index and the 20 bilateral currency rates.<sup>13</sup> The dimension of the covariance matrix in this case is 62×62. We again retain the first 10 out-of-sample principal components, which account for about 70% of the cumulative eigenvalues across the 14 years. We regress the country returns on the 10 out-of-sample principal components and get the adjusted R-squares,  $R^2$ . The impact of exchange rate risk on the estimated R-squares depends on the extent to which currency risks are spanned by the factors extracted from equity returns and the extent to which the unspanned currency risk explains variation in the equity returns. Essentially, perfect spanning of bilateral foreign exchange rate by the common factors imply no effect of currency risk on R-squares. If there is no perfect spanning but the unspanned currency risk's explanatory power to equity returns is small then there will be only marginal effect on PR R-squares. First, we examine the extent to which the currency factors are spanned by the ten principal components extracted from the weekly returns of the global industries and the substitute assets over the entire sample period (1994-2008). Results reported on the online appendix Table A.4 show that some of the variation in the exchange rate factors is explained by the common factors but there is no evidence of perfect spanning. This is especially the case for the global currency factor and the more integrated emerging markets such as Mexico and Korea. For the less integrated emerging markets such as Jordan and Colombia, the local exchange rate factor is only marginally spanned by the common factors. Next, we examine the effect of currency risk on the PR adjusted R-squares by comparing these estimates from the multifactor models with and without currency risk factors.

### **III-3 Results**

Panels A and B of Figure (4) report the plots of cross-country averages of the estimated adjusted R-squares from approaches I and II, respectively. Each plot comprises the annual R-squares with and without currency factors, as well as their differences over the period 1995-2008. Three patterns emerge from the two approaches: (1) the R-squares with and without currency factors are very close; (2) Our sample of emerging markets displays an upward time trend from early 2000s, (3) There were some

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<sup>13</sup> The EM currency index is not used in approach II as the weekly data are only available from 1995.



reversals following the Asian crisis. The mean R-square estimated from approach I increased from 0.4 in 1995 to 0.7 in 2008. Approach II shows a similar upward trend in the mean R-square from 0.3 in 1995 to 0.6 in 2007 and a reversal in 2008 to 0.4. The relatively lower levels of integration measures for approach II are as expected, because of the limited number of substitute assets used.

[Insert Table 5 and Figure 4 about here]

Table 5 presents, for each emerging market, the average R-squares estimated from the two approaches. For each approach, we report the R-squares with and without currency factors as well as their differences. The mean R-square difference estimated from the two approaches is rather small except in a few cases, most notably Indonesia. The estimated difference is positive on average for most countries suggesting a slight downward bias from omitted currency risk. The rankings with/without exchange risk are largely unaffected within each approach. Even across the two approaches, the rankings of the most integrated and least integrated countries are similar- with Mexico, Brazil and Chile being the most integrated and Colombia, Pakistan and Jordan the least integrated. There is slight variation in the rankings of the countries in the middle group. Obviously some of the differences could simply reflect sampling variation in statistical estimation. Thus, the upward trend in market integration and countries rankings are robust to omitted currency factors. Nevertheless, at the country level, there are some variations during crises periods, such as Argentina around its great depression of 1999-2002, the Asian region (Indonesia, Malaysia, South Korea, Taiwan, and Thailand) around the Asian crisis and the recent global financial crisis, and Turkey around its financial crisis of 2001. Plots of the R-squares by country are available from the authors.

## **IV- Conclusion**

Given the importance of market integration and currency risk in international investment and financing decisions, we examine the impact of currency risk on the level and dynamics of market integration based on equilibrium asset pricing models as well as multifactor models. We estimate the international asset pricing model of Chaieb and Errunza (2007) that accounts for segmentation and PPP

deviations and that of Errunza and Losq (1985) which assumes that PPP holds. The estimates of the integration measures based on the CE and EL models should be equal if the global currency premium and the sum of global currency and segflation risk premiums are small. These expectations are confirmed by the empirical results based on the multivariate GARCH-M methodology. We show that the global currency and segflation risk factors do not affect market integration for our sample of twenty EMs except during crises periods. We examine whether deep devaluations occur around the same period as the financial market liberalization, i.e. governments time the liberalization event around currency crisis. The results suggest that in most cases, market integration and currency devaluations do not seem to overlap, thus alleviating the identification concern. Robustness checks on the choice of instruments and parametrization of the prices of risk confirm our main findings.

We also use a measure based on the explanatory power of a multi-factor model as implemented by Pukthuanthong and Roll (2009) and arrive at similar results. We should emphasize that the data sets, the time horizons, data frequency, and empirical methodology used are quite different from those for the equilibrium international asset pricing models. Thus, despite the statistical and economic significance of the currency premiums, the currency risk factors do not significantly affect the degree and dynamics of market integration except during crises periods.

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## Appendix A

Appendix A provides a proof of the Proposition 1 (restated below) and of inequality (4) of the paper.

### Proposition 1

If  $GCP_{t-1} = 0$  and  $(GCP_{t-1} + LCP_{t-1}) = 0$  then  $\Pi_t^{EL} \cong \Pi_t^{CE}$ .

where  $GCP$  is the global currency premium and  $LCP$  is the segflation risk premium defined respectively as:

$$GCP_{t-1} \equiv \sum_{l=MJ,EM} \delta_{l,t-1} cov_{t-1}(r_{DP,t}, q_{lt}), \quad (A.1)$$

$$LCP_{t-1} \equiv \lambda_{q,t-1} cov_{t-1}(r_{I,t}, q_{I,t} | \underline{r}_e) \quad (A.2)$$

**PROOF.** The integration index,  $\Pi_t$ , is defined as

$$\Pi_t = 1 - \frac{var_{t-1}(r_{I,t} | \underline{r}_e)}{var_{t-1}(r_{I,t})}$$

which is equal to the squared correlation between  $r_{I,t}$ , the return on the EM index, and  $r_{DP,t}$ , the diversification portfolio's return,

$$\Pi_t = \frac{cov_{t-1}^2(r_{I,t}, r_{DP,t})}{var_{t-1}(r_{I,t}) var_{t-1}(r_{DP,t})} \quad (A.3)$$

$$= \frac{E_{t-1} [(r_{I,t} - E_t(r_{I,t}))(r_{DP,t} - E_t(r_{DP,t}))]^2}{E_{t-1} [r_{I,t} - E_t(r_{I,t})]^2 E_{t-1} [r_{DP,t} - E_t(r_{DP,t})]^2} \quad (A.4)$$

We want to assess the consequences of the omitted currency risk factors on the measure of integration. Other sources of misspecification, such as errors in the specification of the dynamics of the covariance matrix and/or the dynamics of the prices of risk, are not considered. However, the models are estimated with the same GARCH specification of the dynamics of the covariance matrix of the residuals and with the same parametrization of the prices of risk.

Let  $\Pi_t^{EL}$  denotes the integration index measured using a conditional version of the EL model and  $\Pi_t^{CE}$  is the integration index estimated from a conditional version of the CE model. Though based on the same definition, the two measures will differ since the pricing kernels and hence the expected returns are different. To see this, we write (A.4) as,

$$\Pi_t^M \equiv \frac{E_{t-1} [(r_{I,t} - E_t^M(r_{I,t}))(r_{DP,t} - E_t^M(r_{DP,t}))]^2}{E_{t-1} [(r_{I,t} - E_t^M(r_{I,t}))^2] E_{t-1} [(r_{DP,t} - E_t^M(r_{DP,t}))^2]}, \quad (A.5)$$

where  $\Pi_t^M$  is the integration index obtained from the model  $M = CE, EL$ . According to the EL model, the return generating processes of the EM index and its DP are given by:

$$r_{I,t} = E_t^{EL} [r_{I,t}] + v_{I,t} \quad (\text{A.6})$$

$$r_{DP,t} = E_t^{EL} [r_{DP,t}] + v_{DP,t} \quad (\text{A.7})$$

where  $E_t^{EL} [r_{I,t}]$  and  $E_t^{EL} [r_{DP,t}]$  are, respectively:

$$E_t^{EL} [r_{I,t}] = \delta_{W,t-1} cov_{t-1} (r_{I,t}, r_{W,t}) + \lambda_{I,t-1} var_{t-1} (r_{I,t} | \underline{r}_{e,t}) \quad (\text{A.8})$$

and

$$E_t^{EL} [r_{DP,t}] = \delta_{W,t-1} cov_{t-1} (r_{DP,t}, r_{W,t}) \quad (\text{A.9})$$

According to the CE model, the return generating processes of the EM index and its DP are given by,

$$r_{I,t} = E_t^{CE} [r_{I,t}] + u_{I,t} \quad (\text{A.10})$$

$$r_{DP,t} = E_t^{CE} [r_{DP,t}] + u_{DP,t} \quad (\text{A.11})$$

where

$$\begin{aligned} E_t^{CE} [r_{I,t}] = & \delta_{W,t-1} cov_{t-1} (r_{I,t}, r_{W,t}) + \sum_{l=MJ,EM} \delta_{l,t-1} cov_{t-1} (r_{DP,t}, q_{lt}) + \\ & \lambda_{I,t-1} var_{t-1} (r_{I,t} | \underline{r}_{e,t}) + \lambda_{q,t-1} cov_{t-1} (r_{I,t}, q_{I,t} | \underline{r}_{e,t}) \end{aligned} \quad (\text{A.12})$$

and

$$E_t^{CE} [r_{DP,t}] = \delta_{W,t-1} cov_{t-1} (r_{DP,t}, r_{W,t}) + \sum_{l=MJ,EM} \delta_{l,t-1} cov_{t-1} (r_{DP,t}, q_{lt}) \quad (\text{A.13})$$

We can then write  $E_t^{CE} [r_{I,t}]$ , as the sum of  $E_t^{EL} [r_{I,t}]$ ,  $GCP_{t-1}$  and  $LCP_{t-1}$ . Similarly, we can write  $E_t^{CE} [r_{DP,t}]$  as the sum of  $E_t^{EL} [r_{DP,t}]$  and  $GCP_{t-1}$ . However, these are only approximations since the omitted currency factor from EL model (and potentially other factors from the two models) would result in biased estimates of the prices of risk,

$$E_t^{CE} [r_{I,t}] \cong E_t^{EL} [r_{I,t}] + GCP_{t-1} + LCP_{t-1} \quad (\text{A.14})$$

$$E_t^{CE} [r_{DP,t}] \cong E_t^{EL} [r_{DP,t}] + GCP_{t-1} \quad (\text{A.15})$$

Using equations (A.10), and (A.11) we can write the second moments of the CE model as:

$$E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right) \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right) \right] = E_{t-1} [u_{I,t} u_{DP,t}] \quad (\text{A.16})$$



$$E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right)^2 \right] = E_{t-1} \left[ u_{I,t}^2 \right] \quad (\text{A.17})$$

$$E_{t-1} \left[ \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right)^2 \right] = E_{t-1} \left[ u_{DP,t}^2 \right] \quad (\text{A.18})$$

Now we express the second moments of the EL model as functions of the second moments of the CE model. Using equations (A.6), (A.7), (A.14) and (A.15) we can write:

$$\begin{aligned} & E_{t-1} \left[ \left( r_{I,t} - E_t^{EL} (r_{I,t}) \right) \left( r_{DP,t} - E_t^{EL} (r_{DP,t}) \right) \right] \\ & \cong E_{t-1} \left[ (GCP_{t-1} + LCP_{t-1} + u_{I,t}) (GCP_{t-1} + u_{DP,t}) \right] \\ & = GCP_{t-1} (GCP_{t-1} + LCP_{t-1}) + E_{t-1} [u_{I,t} u_{DP,t}] \\ & = GCP_{t-1} (GCP_{t-1} + LCP_{t-1}) \end{aligned} \quad (\text{A.19})$$

$$+ E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right) \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right) \right] \quad (\text{A.20})$$

Equation (A.19) obtains because the conditional expectations of the error terms are zero. Also we used equation (A.16) to get (A.20).

We do similar calculation for the variance of the returns on the EM index and the diversification portfolio and obtain:

$$E_{t-1} \left[ \left( r_{I,t} - E_t^{EL} (r_{I,t}) \right)^2 \right] \cong (GCP_{t-1} + LCP_{t-1})^2 + E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right)^2 \right] \quad (\text{A.21})$$

and

$$E_{t-1} \left[ \left( r_{DP,t} - E_t^{EL} (r_{DP,t}) \right)^2 \right] \cong GCP_t^2 + E_{t-1} \left[ \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right)^2 \right] \quad (\text{A.22})$$

We can now express  $\Pi_t^{EL}$  as a function of  $\Pi_t^{CE}$  by substituting (A.20), (A.21) and (A.22) in  $\Pi_t^{EL}$  expression given by (A.5) for  $M = EL$ ,

$$\Pi_t^{EL} \cong \frac{\left[ GCP_{t-1} (GCP_{t-1} + LCP_{t-1}) + E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right) \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right) \right] \right]^2}{\left[ \left( (GCP_{t-1} + LCP_{t-1})^2 + E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right)^2 \right] \right) \times \left( GCP_{t-1}^2 + E_{t-1} \left[ \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right)^2 \right] \right) \right]} \quad (\text{A.23})$$

Equation (A.23) shows that if  $GCP_{t-1} = 0$  and  $GCP_{t-1} + LCP_{t-1} = 0$  then  $\Pi_t^{EL} \cong \Pi_t^{CE}$ . Note that  $\Pi_t^{EL}$  could overstate or understate  $\Pi_t^{CE}$  and it is difficult to predict the sign of the difference between the two measures because the currency risk premiums could be negative.

It is clear from the equations (A.21) and (A.22) that:

$$E_{t-1} \left[ \left( r_{I,t} - E_t^{EL} (r_{I,t}) \right)^2 \right] \geq E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right)^2 \right] \quad (\text{A.24})$$

and

$$E_{t-1} \left[ \left( r_{DP,t} - E_t^{EL} (r_{DP,t}) \right)^2 \right] \geq E_{t-1} \left[ \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right)^2 \right] \quad (\text{A.25})$$

Using the inequalities (A.24) and (A.25), we get,

$$\Pi_t^{EL} \leq \left( \frac{\frac{[GCP_{t-1}(GCP_{t-1}+LCP_{t-1})]^2}{E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right)^2 \right] E_{t-1} \left[ \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right)^2 \right]} + \frac{E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right) \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right) \right]^2}{E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right)^2 \right] E_{t-1} \left[ \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right)^2 \right]} + \frac{2GCP_{t-1}(GCP_{t-1}+LCP_{t-1})E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right) \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right) \right]}{E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right)^2 \right] E_{t-1} \left[ \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right)^2 \right]} \right) \quad (\text{A.26})$$

Let  $\Phi_{t-1}$  denotes the non-negative quantity defined as:

$$\Phi_{t-1} \equiv \frac{[GCP_{t-1} (GCP_{t-1} + LCP_{t-1})]^2}{E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right)^2 \right] E_{t-1} \left[ \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right)^2 \right]}, \quad (\text{A.27})$$

and  $\Psi_{t-1}$  is defined as:

$$\Psi_{t-1} \equiv \frac{2GCP_{t-1} (GCP_{t-1} + LCP_{t-1})}{E_{t-1} \left[ \left( r_{I,t} - E_t^{CE} (r_{I,t}) \right) \left( r_{DP,t} - E_t^{CE} (r_{DP,t}) \right) \right]} \quad (\text{A.28})$$

Notice that  $\Psi_{t-1}$  could be either positive or negative depending on the sign of the global currency premium ( $GCP_{t-1}$ ) and sum of global currency and segflation premiums ( $GCP_{t-1} + LCP_{t-1}$ ).

Substituting (A.27) and (A.28) in (A.26), we obtain:

$$\Pi_t^{EL} \leq \Phi_{t-1} + \Pi_t^{CE} (1 + \Psi_{t-1}) \quad (\text{A.29})$$

This is inequality (4) of the paper.

## Appendix B

### Securities for the diversification portfolios

| Country        | Cross-listings                                                                                                                                                                                                                 | Country Funds                          | Country ETF                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------|
| Argentina      | YPF S.A.(USA, 1993/07), BBVA Banco Frances S.A.(USA, 1993/11), Telefonica de Argentina S.A.(USA, 1994/03), Transportadora de Gas Del Sur, S.A.(USA, 1994/11), Metrogas S.A.(USA, 1994/11)                                      | Argentine Investment (1991/10-2001/06) | N.A.                                               |
| Brazil         | Aracruz Celulose (USA, 1992/05), Companhia Brasileira de Distribuicao (USA, 1997/06), Comp. Paranaense de Energia-Copel (USA, 1997/07), Companhia Siderurgica Nacional (USA, 1997/11), Petroleo Brasileiro S.A. (USA, 2000/08) | Brazil Fund (1988/03-2006/06)          | iShares MSCI Brazil Index (10/07/2000)             |
| Chile          | Compania de Telecomunicaciones de Chile (USA, 1990/07), Compania Cervecerias Unidas S.A. (USA, 1992/10), Madeco S.A. (USA, 1993/05), Soc. Quimica y Minera de Chile, S.A. (USA, 1993/09), Enersis S.A. (USA, 1993/10)          | Chile Fund (1989/09)                   | iShares MSCI Chile Investable Mkt Idx (12/11/2007) |
| China          | Sinopec Shanghai Petrochemicals A (USA, 1993/07), Huaneng Power 'H' (USA, 1994/10), Guangshen Railway (USA, 1996/05), China Eastern Airlines (USA, 1997/02), China Southern Airlines (USA, 1997/07)                            | China Fund (1992/07)                   | iShares FTSE China 25 Index Fund (10/05/2004)      |
| Colombia       | Banco Ganadero (USA,1995/01- 2001/03), Bancocolombia (USA, 1995/08)                                                                                                                                                            | N.A.                                   | N.A.                                               |
| Czech Republic | Ceske Radiokom.(Germany, 1996/06), Ceska Sportitel (Germany, 1996/11- 2002/06), Slovakofarma AS Gdr (UK, 1997/12-2003/06)                                                                                                      | Czech Fund (1994/09-1998/02)           | N.A.                                               |
| Hungary        | Ibusz Cert. (Austria, 1990/06), Fotex Cert. (Austria, 1992/11), Zwack Unicum (Germany, 1993/11), Tiszai Vegyi Kom (Germany, 1996/11), OTP Bank (Germany, 1996/11)                                                              | N.A.                                   | N.A.                                               |
| India          | Cesc Ltd (UK, 1996/08), State Bank of India (Germany, 1997/02), Mahanagar Tel.Nigam(UK, 1998/02), Infosys Technologies Limited (USA, 1999/03), Sify Ltd. (USA, 1999/10)                                                        | India Fund (1994/02)                   | N.A.                                               |

## Appendix B

*Continued*

| Country      | Cross-listings                                                                                                                                                                                                               | Country Funds                           | Country ETF                                         |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------|
| Indonesia    | PT Indonesian Satellite Corporation TBK (USA, 1994/10), PT Telekomunikasi Indonesia TBK (USA, 1995/11)                                                                                                                       | Indonesia Fund (1990/03)                | N.A.                                                |
| Jordan       | Arab Potash Co (UK, 1997/12)                                                                                                                                                                                                 | N.A.                                    | N.A.                                                |
| Korea        | Korea Electric Power Corp. (USA, 1994/10), Posco (USA, 1994/10), SK Telecom Co. Ltd. (USA, 1996/06), KT Corporation (USA, 1999/04), Mirae Corp. (USA, 1999/11)                                                               | Korea Fund (1984/08)                    | iShares MSCI South Korea Index (09/05/2000)         |
| Malaysia     | Petaling Tin (UK, 1950/01), Kuala Lumpur Kepong Berhad (UK, 1973/01), Highlands & Lowlands Berhad (UK, 1976/02)                                                                                                              | Malaysia Fund (1987/05)                 | iShares MSCI Malaysia Index (12/03/1996)            |
| Mexico       | Telefonos de Mexico SA de CV - Series A (USA, 1976/01), Tubos de Acero de Mexico S.A. (USA, 1976/01), Vitro S.A. de C.V. (USA, 1991/11), Empresas Ica S.A. de C.V. (USA, 1992/04), Grupo Radio Centro S.A. de C.V. (1993/07) | Mexico Fund (1981/06)                   | iShares MSCI Mexico Investable Mkt Idx (12/03/1996) |
| Pakistan     | N.A.                                                                                                                                                                                                                         | Pakistan Invest. Fund (1993/12-2001/06) | N.A.                                                |
| Peru         | Banco Wiese Limitado (USA, 1994/09-2002/06), Cementos Lima (USA, 1995/03), Compania de Minas Buenaventura (USA, 1996/05), Telefonica del Peru S.A. (USA, 1996/07-2004/03)                                                    | N.A.                                    | N.A.                                                |
| Philippines  | San Carlos Milling Inc. (1962/07-1994/08), Philippines Long Distance Telephone (USA, 1970/01), San Miguel 'B' (USA, 1993/08), JG Summit Holdings (USA, 1995-2001/09), PSI Technologies (USA, 2000/03)                        | First Philippines (1989/11-2003/06)     | N.A.                                                |
| South Africa | Highveld Steel and Vanadium (1981/10), Sasol (1982/04) AngloGold Ashanti (Germany, 1988/09), Barloworldzert (Germany, 1988/09), Absa Group (Germany, 1996/01)                                                                | Southern Africa Fund (1994/02-2004/11)  | iShares MSCI South Africa Index (03/02/2003)        |

## Appendix B

*Continued*

| Country  | Cross-listings                                                                                                                                                                                                                                                                  | Country Funds                     | Country ETF                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------|
| Taiwan   | Macronix International Company Limited (USA, 1996/05 ), Taiwan Semiconductor Manufacturing Co. (USA, 1997/10 ), Advanced Semiconductor Engineering Inc. (USA, 2000/09), United Microelectronics Corporation (USA, 2000/09), Siliconware Precision Ind., Co. Ltd. (USA, 2000/09) | Taiwan Fund (1986/12)             | iShares MSCI Taiwan Index (20/06/2000)              |
| Thailand | Advanced Info. Service (USA, 1993/08), Charoen Pokphand Group (USA, 1994/12), Thai Airways Intl. (Germany, 1997/07), TT&T (Germany, 1998/01), Internet Thailand (Germany, 2002/01)                                                                                              | Thai Fund (1988/02)               | iShares MSCI Thailand Invest Mkt Index (26/03/2008) |
| Turkey   | Tofas Otomobil ADR (UK, 1994/12), Turkcell Iletisim Hizmetleri (USA, 2000/07)                                                                                                                                                                                                   | Turkish Investment Fund (1989/12) | iShares MSCI Turkey Invest Mkt Index (26/03/2008)   |

Appendix B presents the eligible set for each country. Cross-listings are from the four primary depository banks, Citibank, JP Morgan, the Bank of New York Mellon, and Deutsche Bank. Direct and ADR listings are then complemented from the US major exchanges; NYSE, AMEX and NASDAQ. GDR listings are complemented from major world market exchanges, Datastream, as well as the 1998 listings available from Sergei Sarkissian website, <http://web.management.mcgill.ca/Sergei.Sarkissian/>. We report in brackets the inception and end date for the security. All the information was cross-checked and supplemented with the listed company's website. For a full description on the procedure to obtain the ADRs listing please refer to Karolyi (2004). The list of ETFs is from Morningstar, Bloomberg, iShares and CRSP.

**Table 1**  
**Summary statistics for asset returns**

|              | Start date | EM stock excess returns |                          | Real currency returns $q_I$ |                          | Correlation between |          |         |          |
|--------------|------------|-------------------------|--------------------------|-----------------------------|--------------------------|---------------------|----------|---------|----------|
|              |            | Annualized Mean (%)     | Annualized Std. Dev. (%) | Annualized Mean (%)         | Annualized Std. Dev. (%) | EM & $q_I$          | EM & WMP | EM & DP | DP & WMP |
| Argentina    | Jan-88     | 13.34                   | 50.59                    | 1.22                        | 46.39                    | 0.04                | 0.29     | 0.66    | 0.49     |
| Brazil       | Jan-88     | 15.18                   | 54.19                    | 2.60                        | 19.85                    | 0.28                | 0.46     | 0.75    | 0.56     |
| Chile        | Jan-88     | 14.29                   | 24.56                    | 1.48                        | 8.92                     | 0.48                | 0.44     | 0.87    | 0.53     |
| China        | Jan-93     | -3.75                   | 36.73                    | 55.69                       | 22.51                    | 0.00                | 0.48     | 0.87    | 0.54     |
| Colombia     | Jan-93     | 13.40                   | 32.88                    | 2.64                        | 11.60                    | 0.45                | 0.37     | 0.65    | 0.62     |
| Czech Rep.   | Jan-95     | 9.62                    | 30.22                    | 1.68                        | 18.67                    | 0.37                | 0.53     | 0.81    | 0.71     |
| Hungary      | Jan-95     | 9.80                    | 39.74                    | 2.55                        | 13.35                    | 0.55                | 0.68     | 0.92    | 0.76     |
| India        | Jan-93     | 7.43                    | 31.20                    | 1.58                        | 7.59                     | 0.43                | 0.50     | 0.88    | 0.55     |
| Indonesia    | Jan-88     | 7.60                    | 48.23                    | -0.30                       | 25.57                    | 0.53                | 0.36     | 0.75    | 0.52     |
| Jordan       | Jan-88     | -0.46                   | 18.85                    | -0.78                       | 5.49                     | 0.12                | 0.23     | 0.27    | 0.74     |
| Korea        | Jan-88     | 3.32                    | 37.61                    | 0.00                        | 13.53                    | 0.60                | 0.54     | 0.88    | 0.62     |
| Malaysia     | Jan-88     | 4.49                    | 29.25                    | -0.90                       | 8.50                     | 0.46                | 0.44     | 0.71    | 0.70     |
| Mexico       | Jan-88     | 15.61                   | 32.67                    | 2.37                        | 13.79                    | 0.53                | 0.56     | 0.97    | 0.58     |
| Pakistan     | Jan-93     | 2.14                    | 40.75                    | -0.55                       | 6.70                     | 0.16                | 0.13     | 0.56    | 0.27     |
| Peru         | Jan-93     | 15.40                   | 33.45                    | 0.70                        | 5.19                     | 0.26                | 0.49     | 0.84    | 0.54     |
| Philippines  | Jan-88     | 2.92                    | 32.03                    | 0.78                        | 9.05                     | 0.48                | 0.44     | 0.85    | 0.55     |
| South Africa | Jan-93     | 9.30                    | 28.66                    | -0.37                       | 14.87                    | 0.64                | 0.67     | 0.95    | 0.70     |
| Taiwan       | Jan-88     | 2.39                    | 36.91                    | -1.05                       | 5.63                     | 0.23                | 0.42     | 0.69    | 0.59     |
| Thailand     | Jan-88     | 3.94                    | 39.20                    | 0.05                        | 10.42                    | 0.31                | 0.50     | 0.89    | 0.59     |
| Turkey       | Jan-88     | 7.42                    | 56.94                    | 2.80                        | 15.79                    | 0.47                | 0.39     | 0.80    | 0.45     |

Table 1 presents basic statistics for the asset returns on the emerging markets and the bilateral real exchange rates. The emerging equity indices returns are proxied by the MSCI for the twenty emerging markets. All returns are monthly percentage, denominated in US dollar and in excess of the one-month Eurodollar deposit rate. Bilateral real exchange rates are computed from the consumer price indices and nominal exchange rates from IFS. The period is from January 1988 or later to December 2010. For each country, we present the annualized averages and standard deviation, the cross-correlation between each MSCI EM index, the change in real exchange rate ( $q_I$ ), the Diversification Portfolio (DP) and the World Market Portfolio (WMP), as well as the correlation between DP and WMP. The diversification portfolio is constructed as described in Section I.

**Table 2**

**Decomposition of the equity risk premium**

|            | MRP/TRP | CRP/TRP | GCP/TRP | LCP/TRP |              | MRP/TRP | CRP/TRP | GCP/TRP | LCP/TRP |
|------------|---------|---------|---------|---------|--------------|---------|---------|---------|---------|
| Argentina  | 74%     | 26%     | 5%      | 21%     | Korea        | 61%     | 39%     | 18%     | 21%     |
| Brazil     | 47%     | 53%     | 7%      | 45%     | Malaysia     | 68%     | 32%     | 10%     | 22%     |
| Chile      | 88%     | 12%     | 11%     | 1%      | Mexico       | 65%     | 35%     | 17%     | 18%     |
| China      | 61%     | 39%     | 25%     | 15%     | Pakistan     | 56%     | 44%     | 1%      | 42%     |
| Colombia   | 86%     | 14%     | 8%      | 6%      | Peru         | 57%     | 43%     | 10%     | 33%     |
| Czech Rep. | 56%     | 44%     | 4%      | 39%     | Philippines  | 88%     | 12%     | 7%      | 6%      |
| Hungary    | 81%     | 19%     | 14%     | 5%      | South Africa | 74%     | 26%     | 21%     | 5%      |
| India      | 59%     | 41%     | 6%      | 35%     | Taiwan       | 74%     | 26%     | 12%     | 14%     |
| Indonesia  | 85%     | 15%     | 9%      | 6%      | Thailand     | 95%     | 5%      | 4%      | 1%      |
| Jordan     | 51%     | 49%     | 1%      | 48%     | Turkey       | 62%     | 38%     | 8%      | 30%     |

The estimated model in the second-stage is,

$$\left\{ \begin{array}{l} r_{I,t} = \hat{\delta}_{W,t-1} h_{I,W,t} + \hat{\delta}_{MJ,t-1} h_{DP,q_{MJ},t} + \hat{\delta}_{EM,t-1} h_{DP,q_{EM},t} + \lambda_{I,t-1} h_{I,I,t} \left( 1 - \frac{h_{I,DP,t}^2}{h_{I,I,t} h_{DP,DP,t}} \right) \\ \quad + \lambda_{q,t-1} h_{I,q_{I,t}} \left( 1 - \frac{h_{DP,q_{I,t}}}{h_{I,q_{I,t}}} \right) + \varepsilon_{I,t} \\ r_{DP,t} = \hat{\delta}_{W,t-1} h_{DP,W,t} + \hat{\delta}_{MJ,t-1} h_{DP,q_{MJ},t} + \hat{\delta}_{EM,t-1} h_{DP,q_{EM},t} + \varepsilon_{DP,t} \\ q_{I,t} = \hat{\delta}_{W,t-1} h_{q_{I,W},t} + \hat{\delta}_{MJ,t-1} h_{q_{I,q_{MJ},t}} + \hat{\delta}_{EM,t-1} h_{q_{I,q_{EM},t}} + \varepsilon_{q_{I,t}} \end{array} \right.$$

where  $\hat{\delta}_{W,t-1}$ ,  $\hat{\delta}_{MJ,t-1}$ , and  $\hat{\delta}_{EM,t-1}$  are the first-stage estimates of the prices of, respectively, the world market, the MJ real currency and the EM real currency risk factors.

Let  $\varepsilon_t = (\varepsilon_{I,t} \ \varepsilon_{DP,t} \ \varepsilon_{q_{I,t}} \ \hat{\varepsilon}_{W,t} \ \hat{\varepsilon}_{q_{MJ},t} \ \hat{\varepsilon}_{q_{EM},t})'$  be the  $(6 \times 1)$  vector of residuals, where  $\hat{\varepsilon}_{W,t}$ ,  $\hat{\varepsilon}_{q_{MJ},t}$ , and  $\hat{\varepsilon}_{q_{EM},t}$  are the three first-stage estimated residuals of respectively, the world market, the MJ and EM real currency risk factors.  $\varepsilon_t | \Omega_{t-1} \sim N(0, H_t)$  where  $H_t$  is the  $(6 \times 6)$  conditional covariance matrix of the 6 assets conditional on time  $t$  and on the estimated residuals from first stage. The 6 assets are:  $r_{b,t}$ , the country investable index excess return;  $r_{DP,t}$ , the diversification portfolio excess return;  $r_{W,t}$ , the world index excess return;  $q_{MJ,t}$ , the percentage change in the MJ real currency index;  $q_{EM,t}$ , the percentage change in the EM real currency index;  $q_{I,t}$ , the percentage change in the real bilateral exchange rate.  $\lambda_I$  and  $\lambda_q$  are the prices of conditional market risk and segflation risk, respectively. The time-varying prices are estimated with a different set of conditioning information. Specifications for the price of conditional market risk and segflation risk are, respectively, given by:

$$\lambda_{I,t-1} = (k'_I Z_{t-1})^2, \text{ and } \lambda_{q,t-1} = k'_q Z_{t-1}$$

where  $\mathbf{Z}$  is a set of global and local information variables which includes a constant, the U.S. default spread (USDP), the U.S. term structure spread ( $\Delta$ USTP), the world dividend yield in excess of the risk free rate (XWDY), the lagged local equity return, the local dividend yield and the change

in the local exchange rate.

The time-varying conditional covariance  $H_t$  is parameterized as:  $H_t = H_0 * (ii' - aa' - bb') + aa' * \Sigma_{t-1} + bb' * H_{t-1}$ ,

where  $*$  is the Hadamard product,  $a$  and  $b$  are  $(6 \times 1)$  vector of constants,  $i$  is  $(6 \times 1)$  unit vector, and  $\Sigma_{t-1}$  is the matrix of cross error terms,  $\varepsilon_{t-1}\varepsilon'_{t-1}$ .

Country equity indices and the world equity index are from MSCI. The MJ and EM real currency indices are from the Federal Reserve Board. The risk free rate is the one-month Eurodollar rate from Datastream. All returns are denominated in US dollars. Sample is from January 1988 or later to December 2010. The model is estimated by Quasi-Maximum Likelihood.

The table shows the average contribution to the total risk premium (TRP) of the sum of the world and local conditional market risk premiums (MRP), the global real currency risk premium (GCP), the local real risk premiums (LCP), and the real currency risk premium (CRP) which is the sum of the global and local real currency risk premia. The total risk premium is equal to the sum of MRP and CRP. The contributions are measured by the ratio of the average (market or currency) premium in absolute terms to the sum of all the average risk premiums in absolute terms.



**Table 3**  
**Summary statistics for the estimated integration measures and the bias**

|              | Obs. | CF/ETF | # of ADRs | CE model |           | EL model |           | $\Phi_t$ |      |      | $\Psi_t$ |       |      | Mean bias (EL-CE) |
|--------------|------|--------|-----------|----------|-----------|----------|-----------|----------|------|------|----------|-------|------|-------------------|
|              |      |        |           | Mean     | Std. Dev. | Mean     | Std. Dev. | Mean     | Min  | Max  | Mean     | Min   | Max  |                   |
| Argentina    | 276  | 1/0    | 5         | 0.56     | 0.20      | 0.55     | 0.21      | 0.00     | 0.00 | 0.03 | 0.01     | -0.54 | 1.25 | -0.01             |
| Brazil       | 276  | 1/1    | 5         | 0.71     | 0.20      | 0.68     | 0.19      | 0.00     | 0.00 | 0.06 | 0.02     | -0.15 | 0.53 | -0.03             |
| Chile        | 276  | 1/1    | 5         | 0.75     | 0.10      | 0.74     | 0.11      | 0.00     | 0.00 | 0.24 | 0.01     | -0.08 | 1.10 | 0.00              |
| China        | 216  | 1/1    | 5         | 0.65     | 0.17      | 0.66     | 0.16      | 0.00     | 0.00 | 0.09 | 0.01     | -0.49 | 0.68 | 0.00              |
| Colombia     | 216  | 0/0    | 2         | 0.44     | 0.11      | 0.45     | 0.14      | 0.00     | 0.00 | 0.20 | 0.01     | -0.24 | 1.10 | 0.01              |
| Czech Rep.   | 192  | 1/0    | 3         | 0.65     | 0.07      | 0.66     | 0.08      | 0.01     | 0.00 | 0.45 | 0.01     | -1.14 | 1.46 | 0.01              |
| Hungary      | 192  | 0/0    | 5         | 0.86     | 0.05      | 0.85     | 0.02      | 0.00     | 0.00 | 0.22 | 0.04     | -0.01 | 0.97 | -0.01             |
| India        | 216  | 1/0    | 5         | 0.82     | 0.09      | 0.80     | 0.06      | 0.00     | 0.00 | 0.06 | 0.00     | -0.50 | 0.21 | -0.02             |
| Indonesia    | 276  | 1/0    | 2         | 0.59     | 0.11      | 0.60     | 0.12      | 0.00     | 0.00 | 0.11 | 0.02     | -0.20 | 0.78 | 0.01              |
| Jordan       | 276  | 0/0    | 1         | 0.08     | 0.03      | 0.07     | 0.06      | 0.00     | 0.00 | 0.26 | -0.13    | -13.7 | 2.01 | -0.01             |
| Korea        | 276  | 1/1    | 5         | 0.78     | 0.08      | 0.79     | 0.07      | 0.00     | 0.00 | 0.23 | 0.02     | -0.10 | 1.01 | 0.01              |
| Malaysia     | 276  | 1/1    | 3         | 0.55     | 0.10      | 0.59     | 0.14      | 0.01     | 0.00 | 0.56 | 0.04     | -0.04 | 1.90 | 0.05              |
| Mexico       | 276  | 1/1    | 5         | 0.95     | 0.03      | 0.95     | 0.04      | 0.00     | 0.00 | 0.03 | 0.01     | -0.01 | 0.38 | 0.00              |
| Pakistan     | 216  | 1/0    | 0         | 0.33     | 0.18      | 0.32     | 0.18      | 0.04     | 0.00 | 3.24 | -0.28    | -11.5 | 0.81 | -0.02             |
| Peru         | 216  | 0/0    | 4         | 0.71     | 0.10      | 0.70     | 0.13      | 0.00     | 0.00 | 0.07 | 0.01     | -0.08 | 0.57 | -0.01             |
| Philippines  | 276  | 1/0    | 5         | 0.75     | 0.05      | 0.75     | 0.06      | 0.00     | 0.00 | 0.03 | 0.00     | -0.33 | 0.37 | 0.00              |
| South Africa | 216  | 1/1    | 5         | 0.91     | 0.03      | 0.91     | 0.03      | 0.00     | 0.00 | 0.05 | 0.02     | -0.27 | 0.49 | 0.00              |
| Taiwan       | 276  | 1/1    | 5         | 0.55     | 0.21      | 0.58     | 0.22      | 0.00     | 0.00 | 0.01 | 0.01     | -0.16 | 0.26 | 0.02              |
| Thailand     | 276  | 1/1    | 5         | 0.88     | 0.05      | 0.86     | 0.05      | 0.00     | 0.00 | 0.06 | 0.01     | -0.11 | 0.50 | -0.02             |
| Turkey       | 276  | 1/1    | 2         | 0.67     | 0.16      | 0.67     | 0.15      | 0.01     | 0.00 | 0.77 | -0.01    | -1.99 | 0.47 | 0.00              |

For each country, the table presents summary statistics of the integration indices estimated from the CE and EL models detailed in Section 1 of the paper. Specifically, the table reports the mean and standard deviation of the integration measures, as well as the mean, minimum, and maximum of  $\Phi_t$  and  $\Psi_t$  defined as,

$$\Phi_{t-1} \equiv \frac{[GCP_{t-1}(GCP_{t-1} + LCP_{t-1})]^2}{\sigma_{I,t-1}^{2,CE} \sigma_{DP,t-1}^{2,CE}} \quad \text{and} \quad \Psi_{t-1} \equiv \frac{2GCP_{t-1}(GCP_{t-1} + LCP_{t-1})}{cov_{t-1}^{CE}(r_{I,t}, r_{DP,t})}$$

The average bias between the EL and CE integration indices is given by  $E[\pi_t^{EL} - \pi_t^{CE}]$  and is reported in the last column. The sample period is from 1988 or later to 2010.

**Table 4**  
**Relation between integration indices, and foreign exchange regimes**

|              | Obs. | CE model     |               |                |                   | EL model     |               |                |                   |
|--------------|------|--------------|---------------|----------------|-------------------|--------------|---------------|----------------|-------------------|
|              |      | $\alpha_3$   | $\chi^2_{CE}$ | <i>p-value</i> | $\Delta R^2_{CE}$ | $\alpha_3$   | $\chi^2_{EL}$ | <i>p-value</i> | $\Delta R^2_{EL}$ |
| Argentina    | 276  | 13.61        | 3.69          | 0.055          | 2.01%             | <b>12.48</b> | 4.082         | 0.043          | 2.05%             |
| Brazil       | 276  | <b>16.96</b> | 10.23         | 0.001          | 2.30%             | <b>14.00</b> | 5.064         | 0.024          | 1.44%             |
| Chile        | 276  | -0.03        | 0.00          | 0.984          | -0.34%            | -0.49        | 0.136         | 0.713          | -0.31%            |
| China        | 216  | -1.22        | 0.61          | 0.434          | 0.05%             | -1.27        | 0.637         | 0.425          | 0.05%             |
| Colombia     | 216  | 1.05         | 0.20          | 0.655          | -0.33%            | -0.53        | 0.102         | 0.750          | -0.40%            |
| Czech Rep.   | 192  | 1.94         | 0.14          | 0.713          | -0.48%            | 1.85         | 0.154         | 0.695          | -0.46%            |
| Hungary      | 192  | 2.91         | 0.30          | 0.583          | -0.43%            | 12.40        | 0.526         | 0.468          | -0.23%            |
| India        | 216  | -0.04        | 0.00          | 0.979          | -0.47%            | -2.58        | 1.501         | 0.220          | 0.00%             |
| Indonesia    | 276  | -5.53        | 0.90          | 0.342          | 0.18%             | -6.44        | 1.260         | 0.262          | 0.47%             |
| Jordan       | 276  | 3.05         | 1.46          | 0.226          | -0.01%            | -1.13        | 2.593         | 0.107          | -0.07%            |
| Korea        | 276  | -3.01        | 0.28          | 0.594          | -0.10%            | -5.78        | 0.948         | 0.330          | 0.52%             |
| Malaysia     | 276  | 0.90         | 0.15          | 0.703          | -0.23%            | 1.27         | 0.560         | 0.454          | 0.09%             |
| Mexico       | 276  | -6.17        | 0.79          | 0.374          | -0.20%            | -6.69        | 1.036         | 0.309          | -0.08%            |
| Pakistan     | 216  | -1.25        | 1.85          | 0.173          | 0.38%             | -1.09        | 1.818         | 0.178          | 0.34%             |
| Peru         | 216  | 0.87         | 0.54          | 0.463          | -0.15%            | 0.79         | 0.827         | 0.363          | -0.02%            |
| Philippines  | 276  | <b>-5.88</b> | 4.14          | 0.042          | 0.97%             | <b>-4.89</b> | 3.903         | 0.048          | 0.90%             |
| South Africa | 216  | -0.36        | 0.00          | 0.971          | -0.46%            | -3.70        | 0.214         | 0.644          | -0.39%            |
| Taiwan       | 276  | -0.25        | 0.42          | 0.516          | -0.23%            | -0.33        | 0.771         | 0.380          | -0.13%            |
| Thailand     | 276  | -3.23        | 0.31          | 0.578          | -0.19%            | -3.97        | 0.347         | 0.556          | -0.09%            |
| Turkey       | 276  | <b>5.99</b>  | 5.49          | 0.019          | 1.57%             | 2.381        | 0.672         | 0.412          | -0.07%            |

For each country, and each model, the table presents the tests of the two regressions:

$$s_{l,t+1} = \alpha_0 + \alpha_1 s_{l,t} + \alpha_2 i_t + \alpha_3 \hat{\Pi}_t^M + u_{t+1}$$

$$s_{l,t+1} = \alpha_0^* + \alpha_1^* s_{l,t} + \alpha_2^* i_t + u_{t+1}^*$$

where  $s_{l,t}$  is the change in nominal exchange rate versus the US dollar for country  $l$ ,  $i_t$  is the Eurodollar deposit rate, and  $\hat{\Pi}_t^M$  is the estimated degree of integration from model  $M = CE, EL$ . For each model, we report the the coefficient on the estimated integration,  $\alpha_3$ , the  $\chi^2_{CE}$  and  $p$ -value associated with  $\alpha_3$ , as well as the difference between the adjusted  $R^2$ s of the two regressions. All returns are in US dollar term. The sample period is from 1988 or later to 2010.

**Table 5****Average R-squares from the multi-factor model with and without currency factors**

|              | <i>Approach I</i> |                |             | <i>Approach II</i> |                |             |
|--------------|-------------------|----------------|-------------|--------------------|----------------|-------------|
|              | <i>R2</i>         | <i>R2_noFX</i> | $\Delta R2$ | <i>R2</i>          | <i>R2_noFX</i> | $\Delta R2$ |
| Argentina    | 0.76              | 0.75           | 0.01        | 0.43               | 0.47           | -0.04       |
| Brazil       | 0.83              | 0.82           | 0.01        | 0.54               | 0.52           | 0.02        |
| Chile        | 0.81              | 0.79           | 0.02        | 0.64               | 0.66           | -0.02       |
| China        | 0.68              | 0.67           | 0.01        | 0.36               | 0.42           | -0.06       |
| Colombia     | 0.28              | 0.23           | 0.05        | 0.13               | 0.04           | 0.08        |
| Czech Rep.   | 0.43              | 0.38           | 0.05        | 0.21               | 0.22           | -0.01       |
| Hungary      | 0.78              | 0.76           | 0.02        | 0.37               | 0.31           | 0.06        |
| India        | 0.63              | 0.61           | 0.02        | 0.21               | 0.23           | -0.02       |
| Indonesia    | 0.73              | 0.66           | 0.07        | 0.29               | 0.18           | 0.11        |
| Jordan       | 0.00              | 0.02           | -0.02       | -0.03              | -0.05          | 0.02        |
| Korea        | 0.77              | 0.72           | 0.04        | 0.37               | 0.37           | -0.01       |
| Malaysia     | 0.69              | 0.66           | 0.03        | 0.25               | 0.23           | 0.02        |
| Mexico       | 0.92              | 0.90           | 0.01        | 0.68               | 0.68           | 0.00        |
| Pakistan     | 0.23              | 0.25           | -0.02       | 0.05               | 0.05           | 0.01        |
| Peru         | 0.61              | 0.59           | 0.02        | 0.29               | 0.30           | -0.02       |
| Philippines  | 0.53              | 0.45           | 0.07        | 0.28               | 0.26           | 0.02        |
| South Africa | 0.78              | 0.69           | 0.09        | 0.49               | 0.48           | 0.01        |
| Taiwan       | 0.67              | 0.64           | 0.03        | 0.28               | 0.28           | 0.00        |
| Thailand     | 0.65              | 0.62           | 0.02        | 0.34               | 0.35           | -0.02       |
| Turkey       | 0.66              | 0.62           | 0.05        | 0.35               | 0.31           | 0.04        |

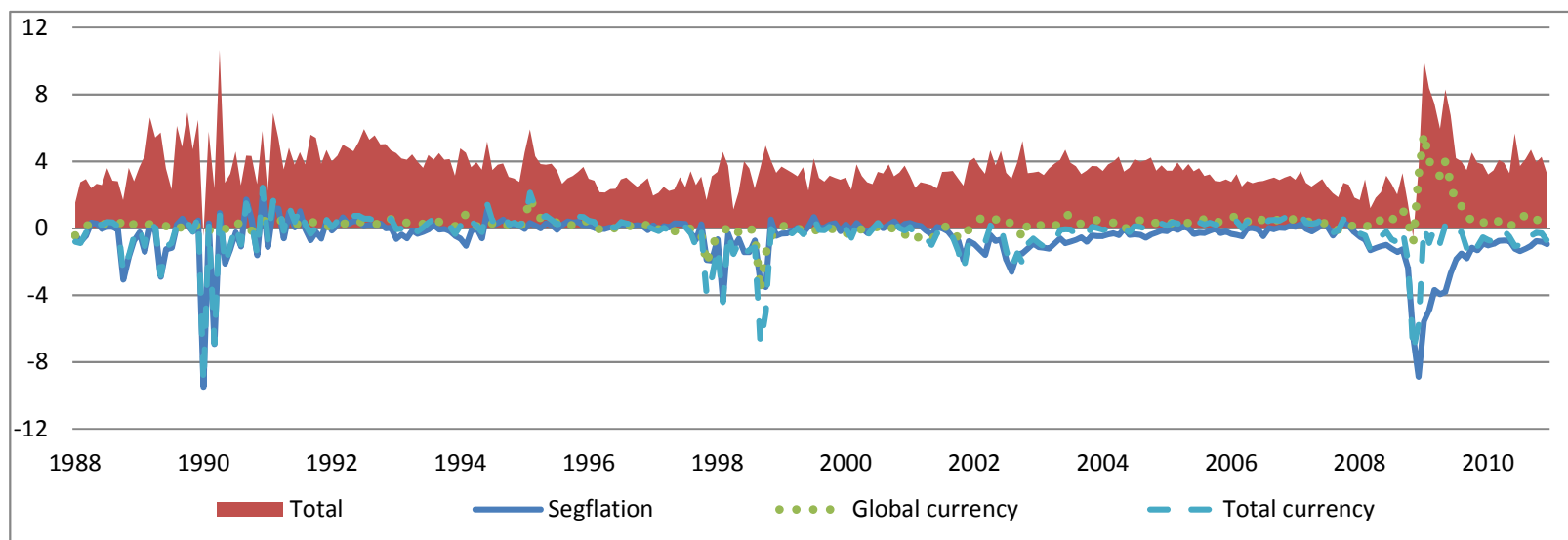
The table presents times-series averages for each emerging market's index of the estimated adjusted annual R-squares with currency factors (*R2*), without currency factors (*R2\_noFX*), as well as their differences ( $\Delta R2 = R2 - R2\_noFX$ ) over the sample period 1995-2008.

The adjusted R-squares are constructed from rolling regressions using weekly returns with a one-year estimation window in the vein of Pukthuanthong and Roll (2009). The explanatory variables of the regressions are constructed in two alternative approaches, (I) and (II).

For approach I, the global factors are estimated by three out-of-sample principal components based on the covariance matrix in the previous calendar year computed with the returns from the level 2 ten global industries. For each country, we also add the CFs, ADRs, ETFs used in constructing the diversification portfolios detailed in Appendix B of the paper. We also add the bilateral exchange rate factors and the nominal currency index factors, MJ and EM to obtain *R2*.

For approach II, the global factors are estimated by ten out-of-sample principal components based on the covariance matrix in the previous calendar year computed with the returns of the level 2 ten global industries and the set of eligible securities, which are present every year from 1994-2008.

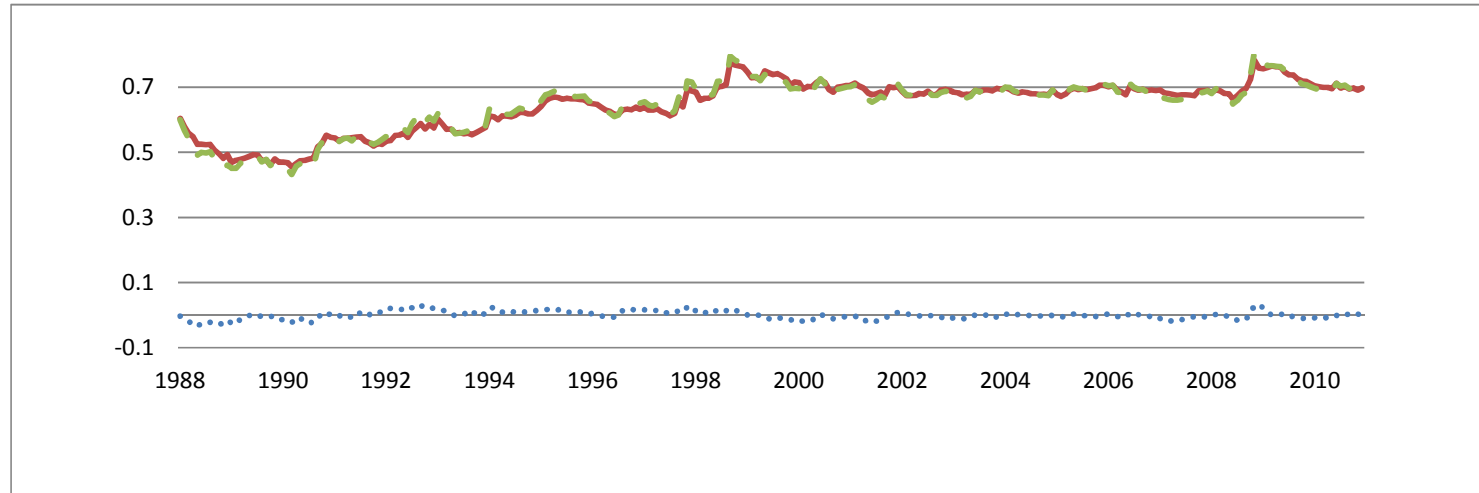
The set of eligible securities include the CFs, ADRs, ETFs of the 20 countries. For *R2*, the covariance matrix also includes the nominal MJ currency index factor and the bilateral exchange rate factors.



**Figure 1**

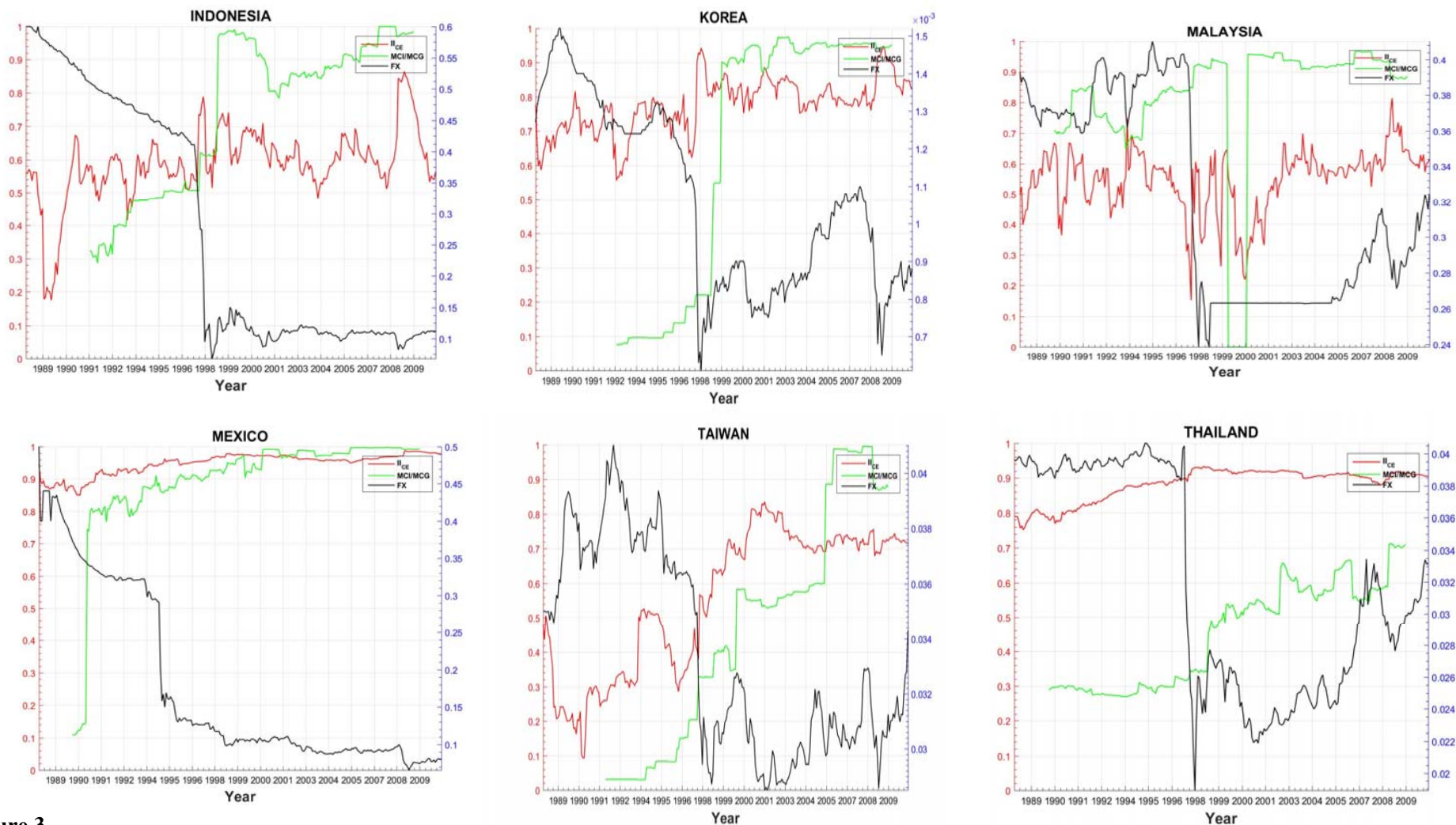
### Time-varying risk premiums

The figure plots the equally-weighted cross-country averages at each point in time of the estimated risk premiums from the CE model presented in section 1 of the paper. The solid line is the segflation risk premium (also termed local currency risk premium), the dotted line is the global real currency risk premium, which is the sum of the MJ and EM real currency risk premiums, the dashed line is the total currency premium, which is the sum of the global real currency risk premium and of segflation risk premium, and the area is the total premium, which is the sum of global and conditional market, global currency and segflation risk premiums. The sample period is monthly from January 1988 to December 2010.



**Figure 2**  
**Time-varying integration indices**

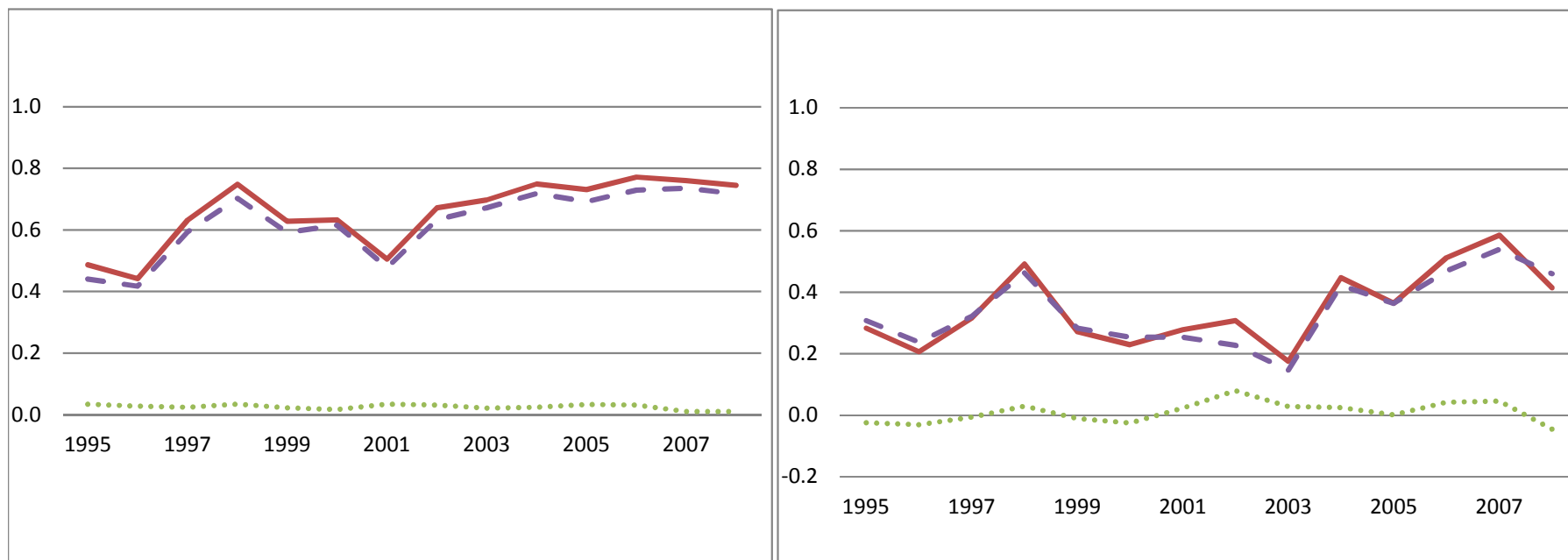
The figure plots the equally-weighted cross-country averages at each point in time of the estimated integration indices for the emerging markets from the CE model (solid line), the EL model (dashed line), and the bias (dotted line). The models detailed in Section I of the paper are estimated allowing for prices and covariances to vary through time. The sample period is monthly from January 1988 to December 2010.



**Figure 3**

### **Time-varying integration indices, nominal FX rates and market investability**

The figure plots the estimated integration indices for some emerging markets estimated from the CE model, the nominal FX rate and the degree of market investability measured as the ratio of investable market cap to total market cap. The right Y-axis display the FX rate and the left Y-axis display the integration index and the degree of investability. The sample period is monthly from January 1988 to December 2010.



**Panel A- Approach (I)**

**Panel B- Approach (II)**

**Figure 4**

### **Sensitivity of the R-squares to the currency factors**

The figures present cross-country averages at each point in time of the estimated adjusted annual R-squares for the IFC investables with global (MJ and EM) and local (bilateral) currency factors ( $R2$  in solid line), without currency factors ( $R2_{noFX}$  in dashed line), as well as their differences ( $\Delta R2 = R2 - R2_{noFX}$  in dotted line) over the sample period 1995-2008.

The adjusted R-squares are constructed from rolling regressions using weekly returns with a one-year estimation window in the vein of Pukthuanthong and Roll (2009). The explanatory variables of the regressions are constructed in two alternative approaches, (I) and (II).

For approach I, the global factors are estimated by three out-of-sample principal components based on the covariance matrix in the previous calendar year computed with the returns from the level 2 ten global industries. For each country, we also add the CFs, ADRs, ETFs used in constructing the diversification portfolios detailed in Appendix B of the paper. We also add the bilateral exchange rate factors and the nominal currency index factors, MJ and EM to obtain  $R2$ . The adjusted R-squares and their differences are plotted in Panel A.

For approach II, the global factors are estimated by ten out-of-sample principal components based on the covariance matrix in the previous calendar year computed with the returns of the level 2 ten global industries and the set of eligible securities, which are present every year from 1994-2008.

The set of eligible securities include the CFs, ADRs, ETFs of the 20 countries. For  $R2$ , the covariance matrix also includes the nominal MJ currency index factor and the bilateral exchange rate factors. The adjusted R-squares and their differences are plotted in Panel B.